GEK-83782

DRIVE SYSTEM INSTRUCTIONS

SILCOMATIC® P L U S 12–PULSE FIRING STATIC KRAEMER ADJUSTABLE S-PEED DRIVE FOR A WOUND ROTOR FAN DRIVE WITH DIRECTOMATIC® PLUS HARDWARE INSTRUCTION MANUAL

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GENERAL ELECTRIC

SILCOMATIC® PLUS
12-PULSE FIRING
STATIC KRAEMER ADJUSTABLE SPEED DRIVE
FOR A WOUND ROTOR FAN DRIVE
WITH
DIRECTO-MATIC® PLUS HARDWARE
INSTRUCTION MANUAL

REQUISITION 541-27816

PLAINS ELECTRIC G & T COOP. INC.

COMBUSTION ENGINEERING

CONTRACT #5780

PURCHASE #2000037

(GEH-83782)

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CHAPTER 1

INTRODUCTION

The purpose of this instruction book is to provide sufficient information to install, maintain and trouble-shoot a SILCOMATIC Plus 12-pulse Kramerstat. These instructions are as general as possible, but they have been designed to supply information that is only required on a specific job. It is written for use by all personnel involved with the power converter (owners, operators, field engineers, electricians, technicians, other maintenance personnel, etc.). Some knowledge of electrical theory and experience in electrical technology is required to implement these instructions. This is not the only information necessary to install a complete system. Information and instructions on the process control equipment, interface cabling, etc. are also required, including the specific power converter elementaries. These instructions do not propose to cover every contingency of installation, maintenance, and trouble-shooting. Should further information be desired or particular problems arise which are not sufficiently covered here, they should be referred to General Electric Company. Recommended changes, additions or deletions to these instructions should be addressed to:

Drive Systems Department
Attn: Power Conversion Engineering
1501 Roanoke Blvd.
Salem, Virginia 24153
Telephone: DIAL COMM 8*278-8672
AC 703-387-8672

CHAPTER 2
SYSTEM DESCRIPTION - POWER CIRCUIT

Functional Description

2-1 The Kraemerstat is a complete rectifier system which converts AC rotor power to DC and inverts this to 60 hertz at line voltage for feedback into the AC system. By varying the firing angle of the inverter, more or less voltage will oppose the DC voltage. As inverter voltage is lowered, higher DC current will flow with consequently higher rotor current. This will increase the torque produced and increase the speed of the rotor. Effectively then, changing the firing angle changes the rotor resistance which changes the speed.

The regulating system consists of an inner current loop and an outer speed loop. These two together provide a smooth adjustment to control the speed over the operating range (50 to 96% of synchronous speed of 900 RPM). Basic components are shown in Figure 2-l.

The Drive has a standby (and starting) operation called the "Standby Mode" of the system whereby the motor is run on resistors. With the starting operation, the motor is started in the usual manner of a wound rotor motor by means of the AC Starting Package (see paragraph 2-6,D for detail). For standby operation, speed is selected by three select pushbuttons at the operation station. During standby operation the smaller the resistance connected to the rotor the more current flows and the greater the available torque (and the greater) the speed.

2-2 Main power disconnect (not supplied as a portion of the Kraemerstat). Primary protection for the wound rotor motor and the converter is provided by a power circuit breaker. Controls are arranged to close and trip the breaker through signals fed to the Kraemerstat. Short circuit (instantaneous overcurrent) and overload protection (time overcurrent) for both the stator and the inverter transformer and the feeder cable from the switchgear to the system trip the breaker directly. In addition, stator differential current will trip the breaker directly.

2-3 Motor - The wound rotor motor is a type M, frame 6380, 8 pole machine. Rated 4500 horsepower, 4000V AC primary, 580A FL stator, 1260V on rotor stand still, 1640A secondary full load, synchronous speed of 900 RPM. A pulse tachometer is provided for speed feedback. Motor differential current transformers (CTs) and overcurrent CTs are mounted in the motor terminal box.

2-4 Diode rectifier - The function of converting the rotor AC power to DC power is performed by the diode rectifier. This is made possible by the ability of the diode to block voltage in one direction and pass it in the other. The diodes are arranged in a three phase bridge configuration. (Figure 2-2).

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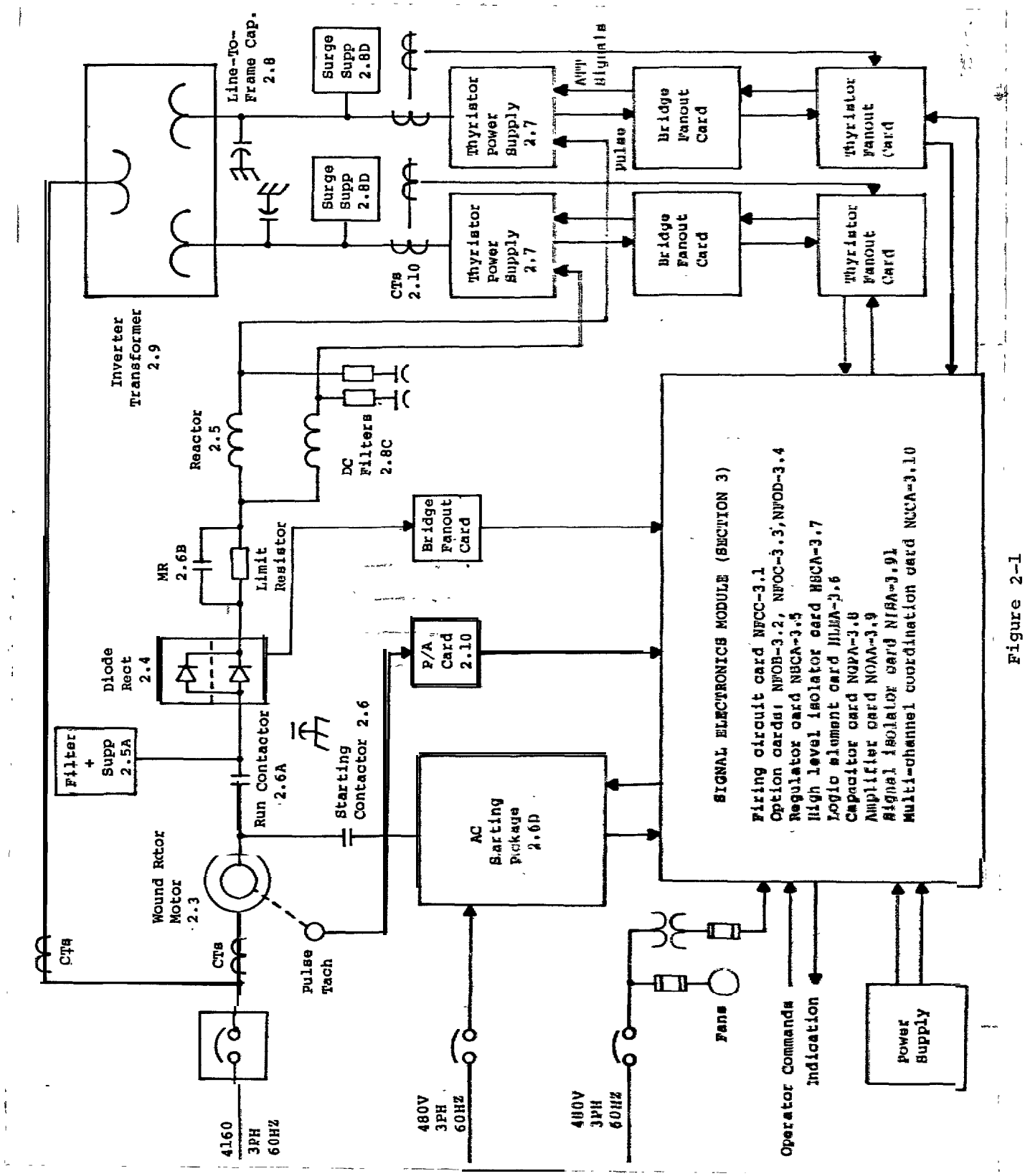


Figure 2-1
Kramerstat Power Converter

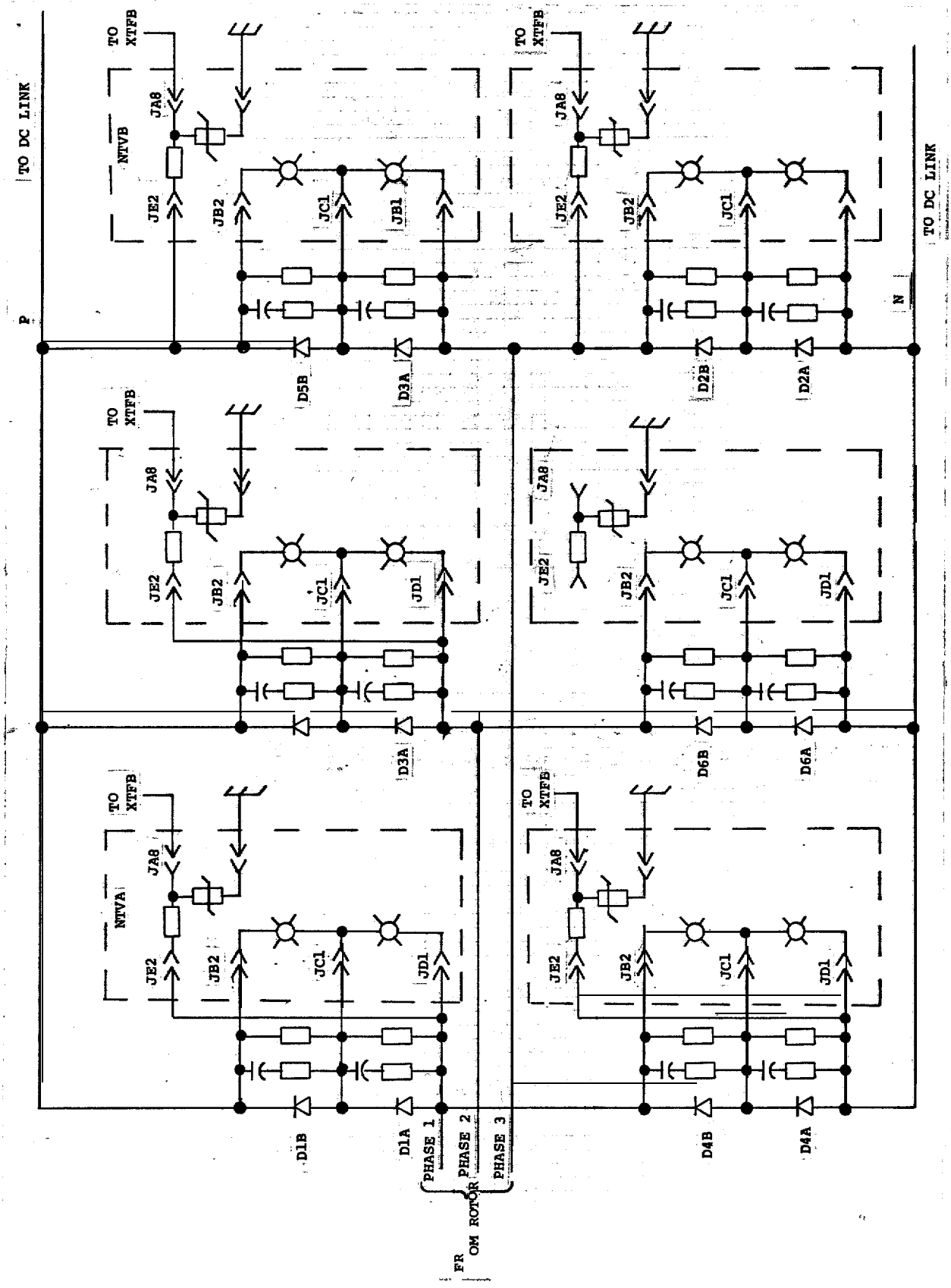


Figure 2-2
Diode Bridge

Each leg of the bridge consists of 2 series connected diodes, heatsink, associated R-C snubber circuits, and a NTVB board containing cell indicating lamps and resistor attenuators. Each diode's PRV exceeds 2.5 times the peak inverse voltage (PREV) imposed by the rotor voltage at minimum running speed (when running in RDS mode). The R-C snubber limits the recovery voltage (in conjunction with the line-to-line filter) to safe values for the cell. The attenuators in conjunction with base mounted resistors form a voltage divider to reduce AC line and DC voltages to values which can be used in the signal electronics module. A second diode bridge - minus attenuators - is used in parallel to handle the large rotor current.

2-5 Reactor - The reactor provides three functions:

A - Limits rate of rise of fault currents so that they do not reach too high a value by the time the running contactor (MR) opens and inserts the current limiting resistor.

B - Attenuates the rotor ripple current to smooth the DC voltage.

C - As the drive increases speed, the rotor voltage drops and the inverter phases forward to maintain current. In order to produce average volts near zero the inverter must fire during the positive half cycle of the inverter transformer voltage. During this period the inverter transformer voltage will drive current through the diode bridge(s).

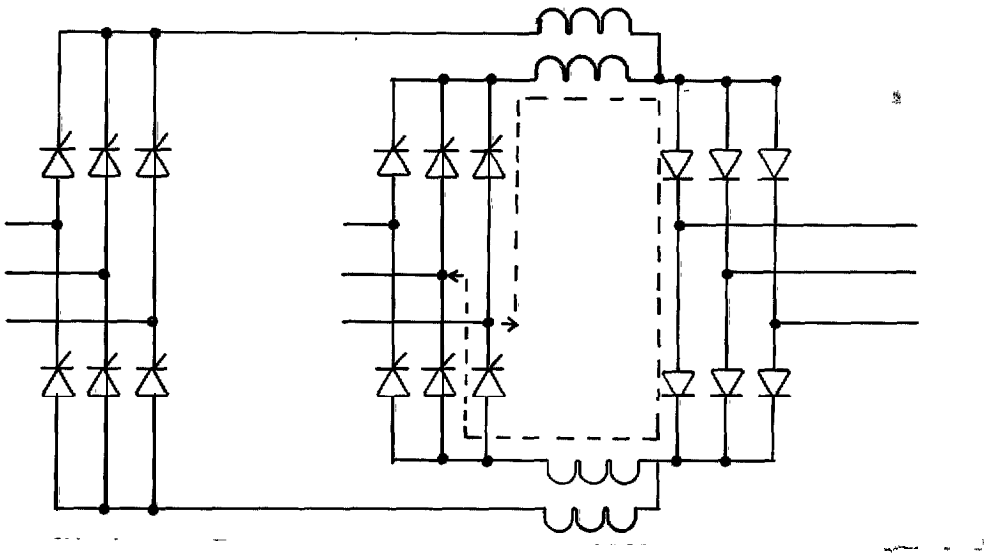


Figure 2-3

The reactor limits this component of inverter current to ten percent of full load DC current. In the case of a 12 pulse circuit, two bifilar winding reactors are used; 1 winding for each bridge (channel) A and B.

2-6 Contactors

A. Running Contactor (Al) - The running contactor connects the rotor to the inverter (diode bridges) after running speed has been reached (min. 50% speed) and the starting contactor(s) (A2) have opened.

B. Limiting Contactor (MR) - The fault current limiting contactor shorts out the fault current limiting resistor when running on the inverter. It limits fault current by opening on faults and reinserts the resistor into the circuit.

C. Starting Contactor (A2) - During starting the starting contactor is closed while the running contactor (Al) is opened. The motor accelerates through the AC starting resistors to running speed. It is coordinated with standby speed contactors (M2 & M3) for accelerating to different preselected speeds. It is also used when the drive is transferred from Kraemerstat mode to standby mode to preselected standby speeds of 50%, 69%, and 79% when so chosen, either manually by the operator or automatically by drive faults.

D. AC Starting Package - The starting package consists of resistors and contactors for accelerating the motor to preselected speeds of 50%, 69%, and 79%. The package is also used during Kraemerstat faults when the drive is transferred to standby mode. At standby mode, different speeds are made available through the use of the standby contactors (M2 & M3) that are closed to reduce the resistance to the rotor's AC current and thus increase the motor speed. If A2 is closed, drive runs at 50% speed, if A2 & M2 close, the drive runs at 69% speed; if A2, M2, and M3 close, the drive runs at 79% speed. All the contactors including MR are latching type with DC operating coils. The coils are connected in series to rectified 480VAC l-Phase. The contactors latch upon closing and must be tripped by energizing other series-connected DC coils.

2-7 Thyristor Power Module. - The function of converting AC power to adjustable voltage DC power is performed by the converter. This is made possible by the ability of the thyristor to block voltage in both directions, while the degree of current flow in one direction is controlled by "gating" the thyristors at adjustable times on the AC voltage wave. Both the basic power circuits and the gating control circuits of the converter are modular, thereby lending the converter to the building-block principle for flexibility and versatility in meeting varied ratings and applications. The basic converter module is a three-phase bridge (Figure 2-4) consisting of six leg (L-N) modules.

A leg module for the SILCOMATIC ® Plus power converter consists of a cell assembly, a pulse amplifier and transformer (PAT) module, and a reactor. Six leg modules are combined to form a full wave bridge. Additional bridges may be paralleled to increase the load current rating of the SILCOMATIC ® Plus converter. Alternately, a second bridge can be connected to second AC source, displaced 30 electrical degrees from the first AC source, to give a 12 pulse system that increases the load current rating of the converter but additionally reduces system harmonics.

A cell assembly consists of the thyristors, heat sinks, and the RC snubber circuit combined into an integral unit. The cell assembly used on these converters is called a series configuration since two thyristors are connected in series, mounted as an assembly, and share a common leg reactor, fuse, PAT, and snubber. The snubber consists of a series reactor and an RC shunt network applied to each cell and conducting path. This arrangement protects the cells against switching surges, ringing, and excessive rates-of-rise of voltage and current during the commutation period. Without this protection, the cells would tend to conduct from changes in voltage, receive excessive changes in current, be overstressed by transient voltages, and tend to switch off after firing due to ringing. Failure would be inevitable, usually from the voltage changes.

Careful visual inspection of these circuits should be made during installation inspection and especially following failure of any cell where the cause is unknown. A cell assembly may be removed from the front of the unit by removing three bolts and unplugging the gate lead connections at the PAT module. Since the cell assembly is an integral unit, field repair should not be attempted. All cell assemblies considered repairable should be returned to the factory for component replacement.

The PAT card located in the leg module contains two parallel circuits for amplifying the firing pulses obtained from the signal electronics and thus firing each of the two cells contained in the leg module. It also maintains electrical isolation (through the use of pulse transformers) between the signal electronics and the power circuits. A secondary function is neon cell indicating lights which normally glow when bridge power is applied but are out when the cell is shorted. Also provided is a LED indicator which lights to indicate presence of pulses to forward cell pulse transformers. Attenuator resistors are also mounted in the PAT assemblies. These resistors are connected to each of the five buses (phase 1, phase 2, phase 3, P, N), however, there is only one attenuator in each PAT. Therefore, five PATS are required to furnish all five signals. This implies that one PAT in a bridge has no attenuator connection. In the 12 pulse systems, both channels (bridges) have their attenuators connected because of the phase shift of the AC signals between the two bridges, (due to transformer), and each operates independently. These attenuators are used in the signal electronics to develop phasing signals, voltage feedback, and cell voltage sensors.

2-8

A - Line-to-Frame Filter Capacitors. These capacitors are used to suppress voltage spikes caused by switching surges. They provide a discharge path between the secondary windings of the transformer and ground.

B - Line-to-Line A-C Filters. These filters are designed to prevent voltage oscillations which would result from switching of the thyristors as commutation progresses from one phase leg to the next. This affect becomes more severe as phase retard increases to the point Of maximum ripple (between 60° and 90° retard, depending on actual inductance of the load circuit), where the switching change of voltage approaches peak line-to-line AC volts ($2E_{L-L}$). Without these filters, overshoot would not only stress the inverse blocking capability of the cells, but ringing could occur sufficiently to switch off the cells after firing.

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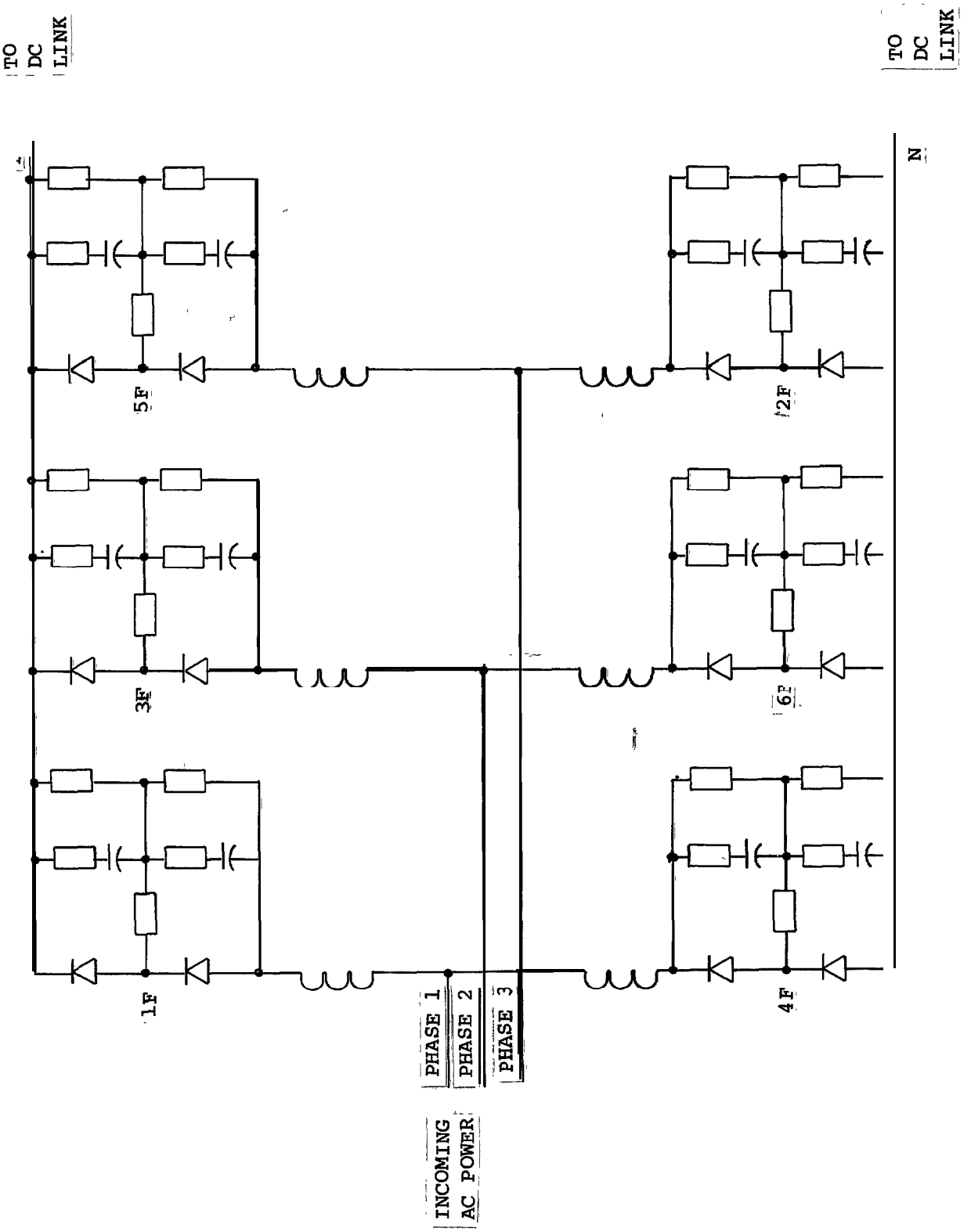


Figure 2-4
Series Cell Bridge Configuration

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C - DC Filters. DC filters, calculated for rated load and inductance values in the DC bus loop, are used for damping purposes within the DC system and to aid commutation when the DC current is discontinuous between successive conduction periods.

D - Surge Suppressor. Surge suppressors, connected line-to-line, provide additional protection against transients caused by "chopping" excitation current on the converter transformer. Their function is to clip line-to-line surges at a value between the peak voltage level of the circuit and the PRV (peak inverse voltage) rating of the thyristors. When the spillover conduction point of the suppressor is reached, the impedance of the device changes instantaneously from a high value to a very low value. It then remains at the low value until the AC line-to-line returns to normal. No adjustments for these devices are provided or intended. They are calculated, fixed values.

2-9 Inverter Transformer. An isolation type transformer couples the rotor electrical power pack into the main AC line through the use of thyristor firing. In this 12 pulse system, the transformer has a dual secondary, one being displaced 30 electrical degrees with respect to the other. This is used to reduce the harmonic currents caused by the thyristors' firing. Each secondary (called a channel or bridge) has its own associated line-to-frame capacitors, surge suppressors, DC reactor, DC filter, fan-out boards, and associated signal electronics. The inverter transformer primary is connected to an isolation switch. The switch also contains the tuning reactors for a 5th harmonic filter. These reactors are coordinated with the power factor capacitors to reduce the 5th, llth, and 13th harmonic currents to acceptable levels. The transformer is protected by overcurrent CTs that trip the incoming switchgear directly. A thermal switch provides overtemperature detection.

2-10 Current transformers. Each channel of the drive has main AC current transformers that are applied on the thyristor AC bus for regulation and protective sensing. The secondaries of these transformers feed the primary of the auxiliary transformers. The secondary of the auxiliary transformers is rectified and fed to the burden resistors mounted on the thyristor fanout card (XTFA). The signal obtained from the burden, resistor is directly proportional to the current in the DC line. This signal is used for regulator feedback as well as fault monitoring.

2-11 Speed Feedback

A - Pulse Tachometer. A pulse tach is provided on the motor. The gear has 96 teeth. It will provide 1440 (1.44KHZ) pulses per second at 900 RPM. Part of the device is mounted on the shaft of the motor, (a metal disc with 96 teeth on the perimeter) and the other part is mounted on the base of the motor close to the shaft, so that when the shaft is turning, the number of pulses produced by the teeth per unit of time is sensed magnetically and is converted to a voltage which is fed back for speed regulation.

2-12 Signal Electronic Modules (SEM)

The 12 pulse Kraemerstat drive has two signal electronic modules (MO1 and M02). MO1 consists of all the electronic circuitry common to both channels (inverter A and B) of the drive. This circuitry is packaged on 18 printed wiring boards, namely 6-NGPA, 6-HLEA, 2-NOAA, NISA, HSCD, NTRA, and NSCA. The functions of MO2 consists of all the electronic circuitry for each individual inverter bridge and the interface with M01. This circuitry is packaged on 10 printed wiring boards, namely 2NFCC, 2NFOB, 2NFOC, 2NFOD, NISA, and NCCA. Schematic information and brief application information is available in the backsheets which can be identified as DS3800XXXX, where xxxx is the 4 letter name for that board. Brief functional descriptions are in Section 3.

2-13 Signal Electronic Power Supply

This base-mounted power supply accepts 110VAC as an input and develops +15, -15, +5 and i-28 volts for use by the signal electronics modules. Circuits included are:

1. Incoming power circuit breaker.
2. Current limit protection on all buses.
3. Remote sensing on +5 volt bus.
4. Potentiometer adjustment of all buses.
5. Power supply status indication.
6. Logic input to signal electronics module - low voltage on any bus is detected.

2-14 Power Factor capacitors - The Hraemerstat Drive (because of the firing of the SCRs) normally has a poor power factor (PF) at low speeds that worsens as the speed increases (current increases; firing angle retards further-worsening the P.F.). Capacitor KVARs (limited by self-excitation value) are added to the drive at the inverter transformer primary to correct the PF to some reasonable value at top speed. The drive's PF now worsens as the motor speed increases.

2-15 Harmonic Filter - Tuning reactors have been coordinated with the PF capacitors to reduce the 5th harmonic currents that are present due to the imbalance of the 2 channels operating in a 12 pulse mode, These are fixed calculated values based on system parameters known at the time.

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CHAPTER 3

KRAEMERSTAT ELECTRONIC MODULE

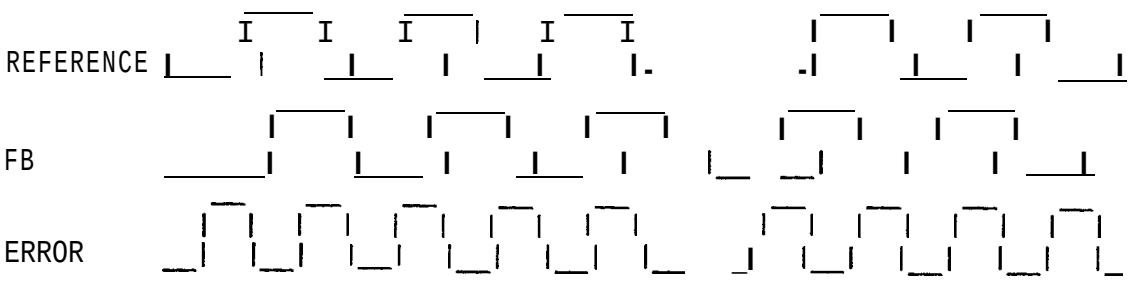
3-l Firing Circuit Board Functions

The basic function of the firing circuit (NFCC card) is to convert the reference obtained from the drive regulator to a proportional DC motor voltage. In addition, it provides converter fault protection and regulator feedback signals.

The firing circuit on the NFCC may be divided into the following functions:

A. Phase locked loop
B. Biased cosine wave generation
C. Cell selection logic
D. Pulse train logic
E. Suicide
F. Feedback signals
G. Logic overcurrent
H. Suppression
I. Overvoltage detection

A. Phase Locked LOOP. (Line Synchronization) Bridge thyristor gate pulses must be synchronized with the AC line to achieve proper firing order and timing. The phase locked loop times the firing of the bridge thyristors with respect to the AC line while providing noise isolation. It isolates cell firing commands from line disturbances such as commutation notches and random spikes generated by other equipment in the plant. Line voltage (phase 1 - phase 3) is obtained using a differential operational amplifier with signals from the option board buffers. Isolation is provided by external resistive attenuators in conjunction with the differential amplifier. This voltage is then filtered (90° lag) to remove commutation notches and noise spikes with the resulting smoothed sinusoid fed to a zero crossing detector which provides a logic "1" output when line voltage is positive and logic "0" when line voltage is negative. The output is then used as the phase lock loop reference. The output of the phase lock loop is a synthesized square wave with a 50% duty cycle. This feedback waveform is fed, along with the reference wave, to a digital phase difference detector (an "Exclusive OR" element), that is, an element having two inputs which produces a "1" output whenever the inputs are different.



Note that the output of the difference detector is a pulse train whose Pulse width is proportional to the difference in phase between the reference and feedback signals. This error signal is applied to an integrating operational amplifier. The result is that its output voltage increases when the reference signal leads the feedback and decreases when the feedback signal lags the reference signal (auxiliary logic steers the error signal to the integrator + and - inputs). The output of the integrator drives a voltage-controlled oscillator which operates at 768 times line frequency at steady-state. The oscillator output is then fed to a binary counter which, along with auxiliary logic, produces a digital signal which is a logic "1" for 384 counts and logic "0" for 384 counts repetitively. This signal is the feedback previously described. Many other oscillator outputs are available. By using these outputs, signals shifted 15, 30, 60 degrees, etc. are obtained. These are used for timing the firing signals. An independent phase-lock monitor operates a card mounted (NFCC) indicating light ("PLL") (normally on). The light goes off to indicate when a differential of 30 degrees or more is sensed.

B. Biased Cosine/Reference Comparison. This produces a linear converter transfer function from analog reference to converter output. An analog cosine wave is generated. The gating signal is then generated by comparing the cosine wave to the analog reference from the drive regulator. Once the fire command signal is generated, control logic increments the cosine wave 60° (lagging) for comparison with the reference so that a fire command signal for the next cell may be generated. In this way, the same cosine wave and comparator may be used to generate firing commands for all cells.

C. cell Selection and control. Basic firing circuit control logic is stored in a digital read-only memory. This logic keeps track of the cell being fired and also selects the next cell to be fired. It also implements inverter limit and several force-fire modes to aid in controlling fault conditions.

D. Pulse Train Firing. Improves drive light load operation (discontinuous current operation). It assures that a cell which is turned off by instantaneous changes in line voltage will be refired immediately. Power Darlington transistors are used in the final output circuit to drive the pulse amplifiers, located in the thyristor bridges. Darlingtons have the advantage of high speed, necessary for transmitting the pulse train firing pulses to the pulse amplifiers. The actual pulse train is produced with a period of 45 microseconds and a 30% duty cycle. The fire command is used to initiate the circuit so that no partial pulses are produced. Information is furnished to route the pulse train to the correct cell pair. Also included in the driver circuitry are inhibit signals to prevent the generation of stray firing signals during power up or down of the power converter.

E. Suicide. A logic signal input will cause the firing circuit to go to a suicide condition; that is, no firing pulses will be generated. If a suicide signal is applied with the drive running, firing pulses will continue to be generated at retard limit until the current goes to zero.

F. Feedback Signals. The source of the current feedback is the burden resistor on the output of a diode bridge connected to the AC current transformers. This signal is then applied to an operational amplifier where it is scaled with the current feedback (CFB) pot. Normal scaling is 7.5 volts at the highest short time rating of the power converter. For example, for a rating of 150% for 60 seconds, the current feedback signal would be scaled to 7.5 volts at 150% current. After scaling, the signal goes through a static switch to generate a bipolar output. voltage feedback is derived using signals from the option card buffers. A differential amplifier measures volts from P to N. The differential OP amp output is then scaled with voltage feedback (VFB) pot with the resultant signal available for regulator use. This is normally adjusted to 4V for rated DC voltage.

G. Logic Overcurrent. Logic overcurrent (LOC) may be thought of as a kind of digital current limit. An analog comparator continuously monitors the burden resistor current feedback. When a preset current level is reached, the output of the comparator causes the firing angle to go to the retard limit, which should cause the current to decrease below the comparator preset. If the reference again increases the current above the preset value, the process will repeat with the current being regulated to approximately the preset value. The LOC setting is tied to the suppression setting, that is, logic overcurrent operation occurs at 80% of the suppression value. Normally suppression will be set at 1.4 times the highest power converter short time rating. This results in a LOC setting at 1.1 times the highest short time rating.

H. Suppression. An analog comparator monitors the current feedback signal obtained from the burden resistor. When current exceeds 1.4 x the highest short time rating value, a flip-flop is latched, sealing in the fault condition and turning on the "SUP" indicating LED. The firing circuit logic signal forces the firing angle to the retard limit under the constraints of the inverter limit circuitry, until zero current is attained. The forced firing is released and drive reverts to normal operation. If a second suppression occurs within 2 seconds of the first, after zero current is attained, the firing pulses are removed and the drive is transferred to standby.

I. Overvoltage Detection. voltage feedback is fed to a comparator and compared with a fixed reference. If the peak of "VFB" is greater than 8.8 volts, 10V becomes a logic one. This signal is latched, indicated by OV (a red LED) which becomes luminous and is transmitted to the firing circuit. The firing circuit fires the diametric cell of the positive cell selected. The signal (IDFO) is also transmitted to the fault circuit.

3-2 NFOB

Buffers. The buffer amplifiers are used to electrically isolate and unload the attenuator signals. Attenuator signals for Phases 1, 2, and 3 and the positive (P) and negative (N) DC buses come from connections existing in the master bridge. Connections are between these buses and five out of the six pulse amplifier and transformer (PAT) cards through dropping resistors on the PAT end. These signals are then transmitted through 10-pin ribbon cables to the bridge fanout card(s) (XBFA) and onto the thyristor fanout card (XTFA). The daughter board resistors on the XTFA card scale the signals and feed them to the buffers on NFOB. The buffer outputs are then fed back to the firing circuit board (NFCC).

****
* WARNING *
****

NEVER APPLY BRIDGE POWER WITH CONVERTER IN TEST MODE: THE PURPOSE OF THE TEST OSCILLATOR IS TO PROVIDE A REFERENCE SIGNAL TO THE FIRING CIRCUIT PHASE LOCK LOOP SO THAT CORRECT CONVERTER OPERATION MAY BE INSURED WITHOUT APPLYING BRIDGE POWER.

3-3 NFOC Option Board Functions

A. Inversion Fault Detector. Commutation on a 3 phase bridge occurs between cells connected to the positive bus and between cells connected to the negative bus. Cell one commutates to cell 3 which, in turn, commutates to cell 5. Cell 5 commutates to cell 1.

On the negative side, cell 4 commutates to cell 6, 6 to 2, and 2 to 4. See Figure 3-l.

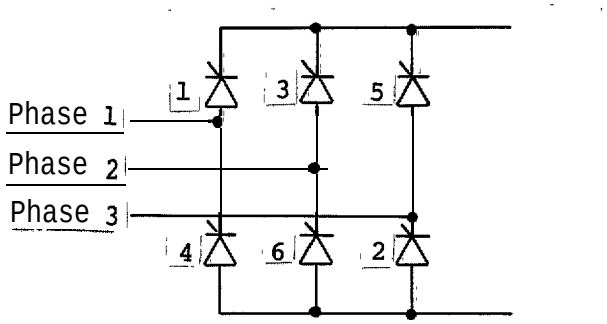


Figure 3-l

Cells 1 and 2 conduct on PHl to PH3 voltage. When cell 3 is fired, commutation will take place from V1-3 to V2-3. Commutation must be complete while voltage 2-3 is more positive than 1-3. See Figure 3-2.

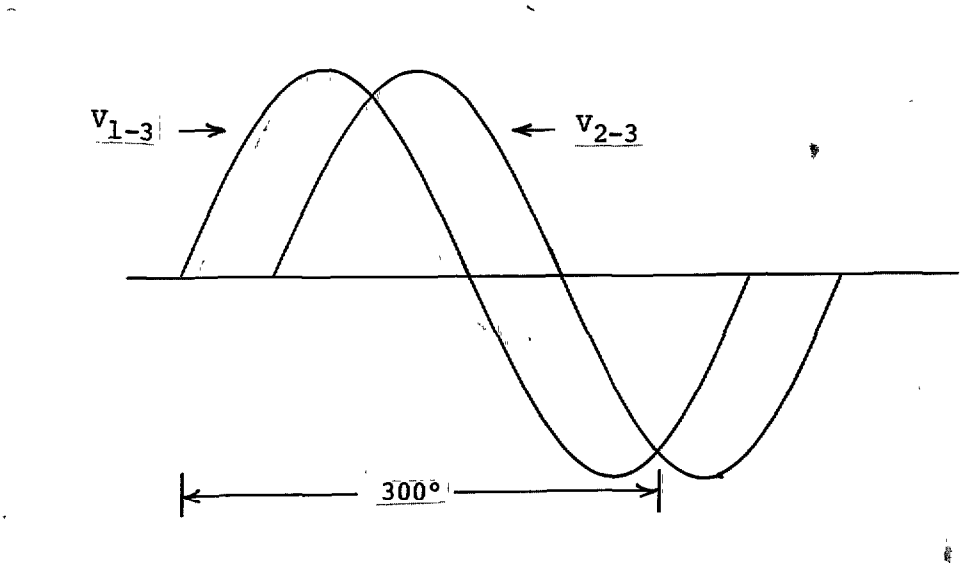


Figure 3-2

This point occurs when vl-3 is 300° from its positive-going zero point. If cell one is conducting after this point, commutation will not be completed and an inversion fault exists.

The 3 phases a'nd the positive and negative bus voltages to neutral are obtained from the NFOB board. Each phase is compared to the positive voltage and the negative voltage separately. Figure 3-3 shows the circuit for cell #1, and the voltage wave shapes for PBUS-N and the phase-N voltages.

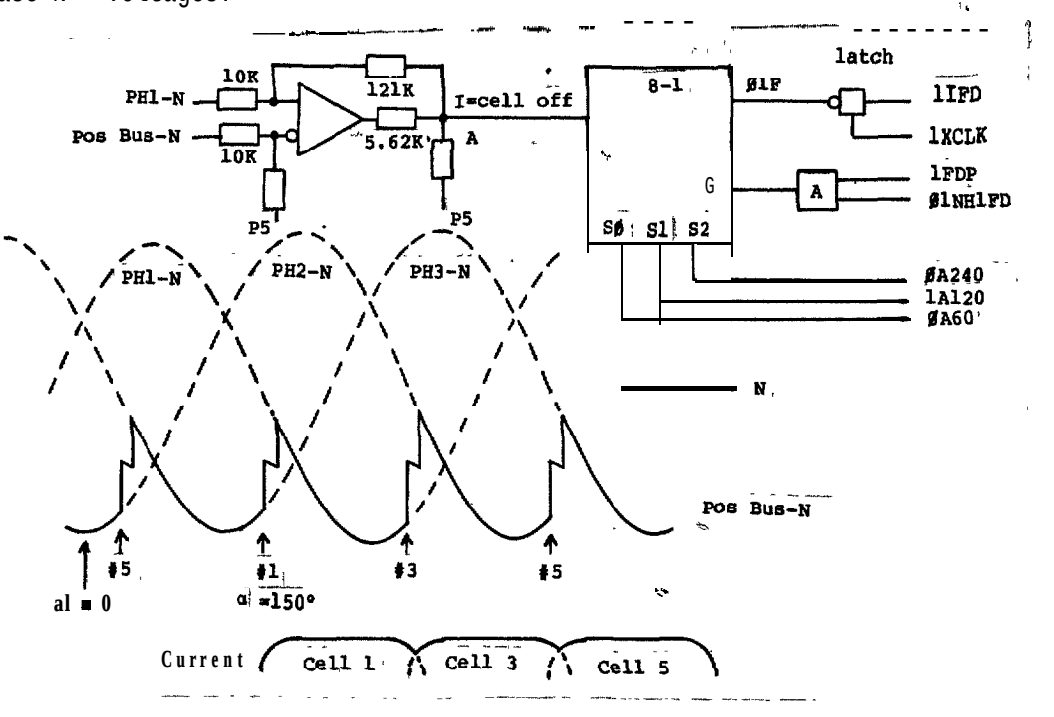


Figure 3-3

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Voltage wave shapes POS BUS-N and phase voltages = 150° (wave shapes caused by cells 2-4 and 6 omitted for clarity). While cell one is conducting (= 150 to = 270 + COM), the PBUS-N and phase 1-N voltages are the same. The P5 bias on the inverting input of the comparator will drive the comparator output to -15v. Point A will be a logic 0. If commutation is complete by = 300° line 1-N will become more positive than PBUS-N. The output of the comparator will be driven to positive 15V. Point "A" will be a logic 1. The outputs of the comparators are connected to a data select multiplexer which selects each cell during its = 300 to 360°. Information for the selection is obtained from the NFCC. Figure 3-4 shows the wave shapes of these signals. If cell one is conducting during the 300°- 360° period, a "zero" will be transmitted to the multiplexer output providing the gate is a zero. This sets the latch in conjunction with the NFCB clock pulse. 1IFD becomes a logic one. 1FD is fed to the NFCC which omits pulse 4 and advances 60° to pulse 5 in an attempt to pull out of the fault.

The NFCB sends a "1" signal (1IFDP) to the logic element feeding the multiplexer gate starting 15° after Alpha equal to 300° lasting for 15°. This permits the multiplexer to transmit the selected cell state to the latch, provided the firing circuit has detected presence of current (IONHIFD = 1). When IFDP returns to a zero, it resets the latch by taking the gate high in preparation for the next cell firing. IFD is also fed to the fault circuit (see P7-6E2).

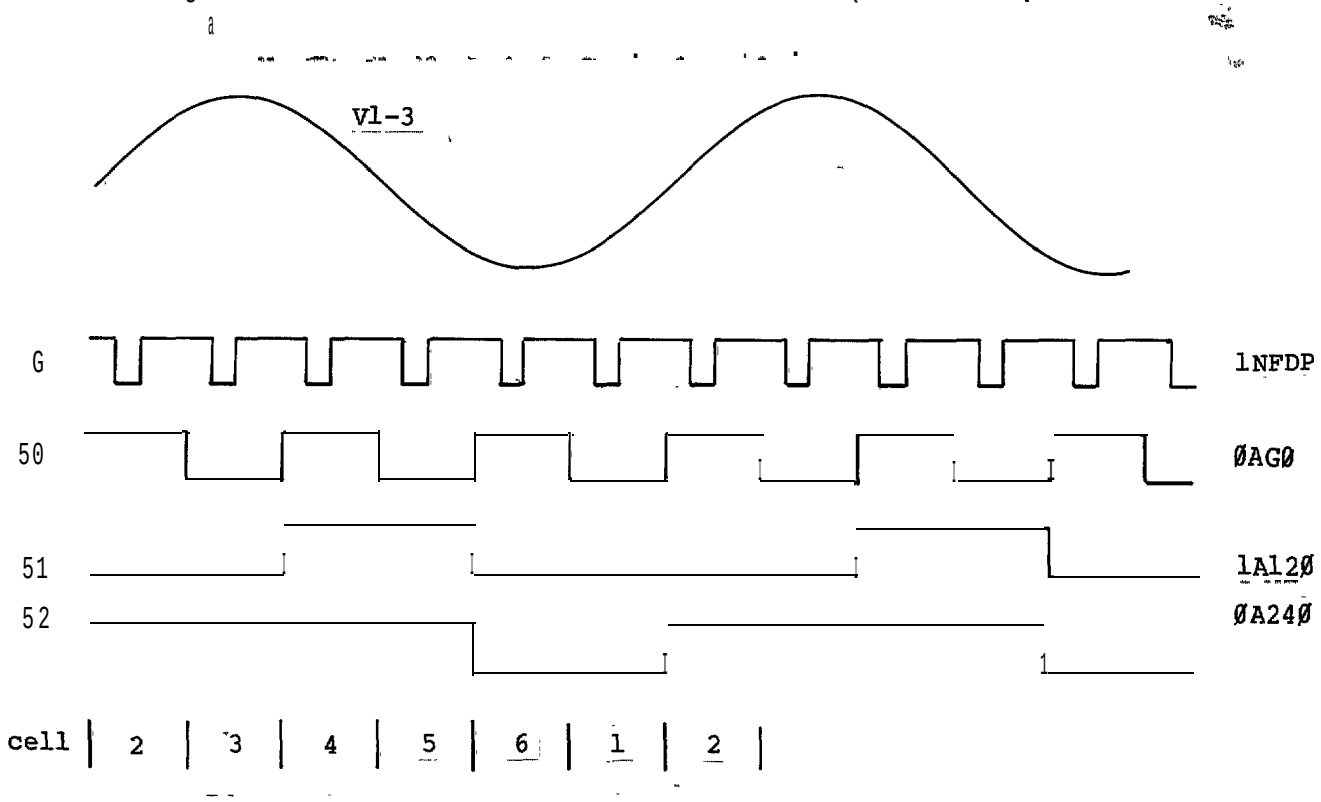


Figure 3-4

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Multiplexer				Function Table	
S2	S1	SO	G	Output	
X	X	X	H	Z	X
L	L	L	L	0	1
L	L	H	L	1	6
L	H	L	L	2	Not used
L	H	H	L	3	Not used
H	L	L	L	4	3
H	L	H	L	5	2
H	H	H	L	6	5
H	H	H	L	7	4

Figure 3-4 (Continued)

B. Thermal Model. The purpose of the thermal model is to protect the power converter against long-time cyclic-type overloads. Whereas the suppression circuitry monitors the instantaneous value of current, the thermal model observes the long time (5-120 seconds) heating effects of the load current. The thermal models are preconditioned to rated ambient temperature. Ambient temperature is defined as the converter-rated ambient temperature. This is usually either 40°C or 50°C. An analog computer-like model simulates the thermal characteristics of the thyristor and heatsink. using the current feedback signal (CFB) as an input, the model calculates the instantaneous cell junction temperature. The output corresponding to the temperature of the forward thermal model is compared with a fixed 110°C reference. When a value of junction temperature exceeding the reference temperature is calculated, the CTMP lamp on the NFOB comes on; the same signal is ANDed with the fan loss signal and fed into a timer. The output of the timer, after a time delay depending on OZCXDl, is latched and fed into the fault signal. In order to fully use the cell capabilities, the ambient input to the thermal model is adjusted to 15° below the actual ambient temperature (in this case ambient temperature is 40°C) so that CTMP will indicate a cell temperature of 125°C (125°C represents the highest "safe" junction temperature). The timer is made practically instantaneous by Connecting OZCDl to the NFCC timer. The latch is reset by the NFCC suppression reset switch.

3-4 NFOD Option Board Functions

A. AC Voltage Compensation. The three phase voltages obtained from the NFOB board are combined to produce line voltages l-2, 2-3, and 3-l. The negative portion of these voltages is eliminated. The resultant positive halfs are added through a suitable gain to produce a bias voltage VCOMP. This voltage, which will increase as line voltage increases, is fed into the bias of the NFCB. An increase in bias phases the rectifier off to maintain constant output voltage.

B. Phase Loss/Sequence. The purpose of the phase loss/phase sequence detector circuit is to monitor the 3 phase voltages applied to the bridge for proper l-2-3 phase sequence and to check for single phasing. If an improper phase condition is sensed, the fault signal causes the firing circuit to be suicided. The fault is indicated by "PH SEQ" LED on the NFOD being on.

C. AC Fault Logic. The purpose of this function is to provide a means of opening the switchgear feeding the Kramerstat in the event AC fault current is not interrupted by the suppression circuit. If an overcurrent persists for longer than 27 milliseconds an output 1ACFLT is sent to the fault logic circuit (PAR 1-6E2) which will open the primary feeder.

3-5 NSCA Functions

A. Speed Regulator. The speed regulator is adjusted for a crossover of about 2.5 radians. A lead in the speed feedback (C12A, 12B X R20) with C12A + C12B = 4.5 MFD and R20 fixed at 100K gives good response. This lead is compensated by a lead in the feedback of the FWD gain consisting of c16 (R54 + 47.5K). Both of these are set at about 2.5 radians.

$$\frac{1}{c16 \times (R54+47.5K)} = \frac{1}{0.68\ MFD \times 609.5K} = 2.41 \text{ and } \frac{1}{C12 \times 100K} =$$

2.22 radians. The gain in the circuit is set by the value of c16 and may be adjusted by R3. With R3 set in the center, the crossover of the speed regulator should be about 2.5 radians. The response to a small step change in speed reference should be nearly critically damped with a 0.4 second response.

B. Current limit on the inverter is set with R4 on the NSCA board and is set at 7.5 volts corresponding to 150% rated current.

C. Logic is provided on the NSCA to suicide the speed regulator and firing circuit from signals during operation.

1. Pin 80 SUAX 1 suicides speed regulator - not used
2. Pin 74 SUA 1 suicides speed regulator and firing circuit 0 unsuicides firing circuit and permissive to unsuicide speed regulator

3. Pin 76 OSUS 0 suicides speed regulator
   1 suicides S.R. if SUA = 0

Further logic is provided to accommodate testing.

1. Pin 67 OSUA 0 suicides speed regulator and firing circuit
2. Pin 66 ORLA 0 unsuicides all regulators
3. Pin 64 ORLT 0 unsuicides firing circuit
4. Pin 59 OSUSX 0 unsuicides firing circuit

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D. A filter for voltage feedback is provided to remove high frequency ripple.

3-6 High level Isolator Card HSCD. Remote and local switches are connected to the input driven by 105V DC. Light isolation is provided and inverts the input (a closed contact gives a logic 0).

3-7 Logic Element Cards HLEA.

1. Six AND/OR gates are provided on each card. A true and not true output are provided.
2. Two four-input AND gates are provided per card.
3. Three inverter gates are provided per card.
4. Three relay drivers per card are provided. A logic zero provides 28V to drive a relay. An inverting logic output is also provided.

3-8 Gain Programming Card NGPA. The following functions are provided:

1. Two general purpose operational amplifiers.
2. Two three-input NAND gates.
3. Two analog (FET) switches, a high logic of 2 volts closes the switch. A low logic of 0.8V opens the switch. Input current is 0.8 MA for a low and 40 microamperes for a high (2 low power Schottky loads). The switch is good for 60 MA continuous current. Care should be exercised when working with the switches since they are electrostatic sensitive and can be damaged by voltages in excess of ±15 volts with respect to ACOM.

4. Two general purpose dual comparators. The input voltage is compared to an adjustable reference. When the input is less positive than the reference and more positive than the inverted reference, a logic "0" is obtained on the non-inverting output. When the input is more positive than the reference or more negative than the inverted reference a logic one output is obtained on the non-inverting output. One comparator provides separate inputs for positive and negative voltages.

5. One relay driver which provides 28v at 350 MA is provided. A logic zero on the input activates the output.

6. One time delay circuit. When both inputs (or the last of either) go from a low to a high, the non-inverting output changes from low to high after a time delay. The time delay is adjustable from 0-10 seconds.

7. Analog Multiplier/Divider. The two analog inputs PX-NX and PY-NY are multiplied together and appear at the VO and NZ outputs. The input and output ranges are ±10V full scale. Inputs above 12V may damage the chip. The gain expression for the multiplier is:

$$VO \text{ (to COM)} = \frac{(PXIN-NXIN)\ (PYIN-NYIN)}{10} + NZIN$$

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Basic accuracy is up to 40MV constant output offset at 25°C. The output offset remains constant regardless of the input.

8. A daughter board is provided for variable components.

3-9 Amplifier Cards NOAA. Each card contains the following components:

1. 4 inverting general purpose operational amplifiers
2. 2 differential operational amplifiers
3. An adjustable clamp for use with (1)
4. A daughter board for variable components.

3-10 Signal Isolator NISA. Each card contains:

1. 8 digital line drivers
2. 8 digital line receivers. The line receivers are optically coupled for maximum isolation.
3. 2 differential operational amplifiers for receiving analog signals.
4. 1 variable gain operational amplifier
5. 1 OR GATE.

The digital line drivers are designed to operate line receivers on a remote NISA card. Similarly the line receivers should be driven from line drivers on a remote NISA card.

3-11 NCCA card

1. Averagers. Two amplifiers, one for CFB and one for VFB, average the feedback signals from as many as 4 bridges (2 used in this case). In addition a current balancing network insures that each bridge shares the inverter load equally. Included are switches for eliminating selected currents from the balance network.

2. Fault Combination. OR-gates are used to reduce the number of fault signals by combining the various common fault signals from the bridges to give a resultant fault indication.

3-12 Thyristor Fan-out Board (XTFA)

The XTFA board interfaces the power bridges to the signal electronic module. The board is base-mounted in the control compartment. Inputs and outputs are as follows:

Connector Designation	cable/Signal Description
JA	34-pin ribbon cable to the front of the firing circuit board (NFCC). This cable carries firing pulses from the NFCC and phasing signals to NFCC.
JB	10-pin ribbon cable to the XBFA card, mounted over the bridge compartments. It carries attenuator signals from the thyristor bridge.

Connector Designation	Cable/Signal Description
JC	26-pin ribbon cable to the XBFA board, mounted over the bridge compartment. It carries 28 VDC power and firing signals to the bridge-mounted pulse amplifier.
JD	Not used here.
JE	26-pin ribbon cable to the front of the firing circuit option card (NFOB). It carries attenuator signals to the NFOB and buffered outputs back to NFCC.
JF	26-pin ribbon cable connector, used for the test mode only. TO initiate the test mode, the 26-pin ribbon cable from the NFOB card is removed from JE and inserted in JF.
JC	10-pin discrete wire connector from the signal electronics power supply. It carries 28V and DCOM. The power is then distributed to the pulse transformers.
JH	6-pin discrete wire connector from auxiliary current transformers. A diode bridge on the XTFA card rectifies the CT signal. The CT burden resistors are also mounted on the XTFA daughter board.

The attenuator scaling resistors function as a signal distribution center. Selectable resistors are located on the XTFA daughter board and are factory selected for proper attenuation for the rated voltage.

3-13 Bridge Fan-out Board (XBFA)

There are two XBFA boards. One is for the diode bridge and the other is for the SCR bridge. The one for the diode bridge carries the attenuator connections to the signal electronic module by means of the ribbon cable. The one for the thyristor bridge receives the 28 volt power and firing signals from the XTFA board through a 26-pin ribbon connector. Included are 10-pin ribbon cables used to distribute the firing pulses to the pulse amplifiers. One 10-pin ribbon cable per pulse amplifier is used. Another 10-pin ribbon connector (JH) carries the attenuator connections to the XTFA board.

27 287 blank



CHAPTER 4

OPERATIONAL SEQUENCE OF STARTING, TRANSFERRING AND STOPPING

4-l General

This drive has two modes of operation: standby and Kraemerstat. The drive must start in standby mode by means of the AC resistor package. The minimum speed is 50% (450 RPM). When it is in standby mode, the drive can run at any one of three preselected speeds: 50%, 69%, 79%, which are selected by pushbuttons located at the operator station. When the Kraemerstat mode is desired, the drive must be in standby mode before switching mode. In Kraemerstat mode, speed can be varied from 50% speed to 96% speed by means of the reference from the speed potentiometer or the auto-reference from the process regulator.

Whatever mode is chosen, the blowers for that mode must be on prior to operating in that mode. Also, there is available on the KDS drive a lockout switch. When moved to either standby or KDS, it prevents operation in that mode. While in normal, operation is in either mode and automatic transfers (due to KDS faults) are allowed.

4-2 Start Sequencing

A. Initial conditions include:

No building overtemperature,
No AC fault or switchgear lockout,
Starting contactor A2 is open,
Kds lockout switch is in normal or KDS
Main contactors are correct mechanically and ICS picks-up

B. Depress STDBY BLWR (standby blower) START pushbutton (PB). STDBY RDY (standby ready) light and STDBY BLOWR START PB light are on.

C. Depress one of the % spd req PB (50,69,79).

1. The requested speed relay P-U (picks-up), the SPD REQ PB light is on.
2. The starting contactor A2 closes and STDBY RDY light is off, after 0.5 second ICS drops out.
3. The main breaker closes, OV picks-up, the STOP light is off, and the drive accelerates to 50% speed, OV drops out and 50% timer starts.

4A. If 69% or 79% SPD were requested, M2 closes (after 50% timer timed out), drive accelerates to 69% speed and 69% timer starts.

4B. If 79% spd was requested, M3 closes (after 69% timer times out), drive accelerates to 79% spd and 79% timer starts.

Note: For 3, 4A, 4B when the speed requested is reached, the % SPD light is on.

4-3 Stopping From Standby

A. Depress stop PB

1. Main breaker (M) trips and STOP light comes on.

2. Starting contactor (A2) trips, motor coasts down, and STDBY READY light is on.

4-4 Run in KDS Mode
Initial conditions include:

A. The lockout switch in NORMAL position and the KDS blowers are on. After reaching 50% spd via the SPD REQ PBS, the KDS READY light will be on.

B. Depress KDS Request PB

1. KDS REQ and IRUN pick up and KDS REQ light is on.

2. MR picks-up, bypassing the current limit resistor.

3. The firing circuit (FC) is unsuicided and a 90° retard reference is applied and a speed regulator (SR) preconditioning (50%) is applied.

4. KDS contactor (Al) closes, KDS takes current to reach 20% FL, and KDS READY light is out. The requested spd relay drops out and REQed SPD LITE is out.

5. Starting contactor (A2) trips, the SR is released, the 90° retard and preconditioning are removed, the remote ref is applied, the STANDBY READY and KDS ON lights are on and the KDS REQ light is off.

6. Drive follows the remote speed reference.

C. The lockout switch may now be moved to the standby position to eliminate any transfer to the standby mode due either to manual attempts by the operator or to automatic transfers requested by KDS due to faults.

4-5 Stopping In KDS Mode

A. Depress Stop Pushbutton

1. MR drops out inserting the current limiting resistor, the drive suicides and fires diametric cells.

2. The main breaker (M) trips, the diametric firing stops and the STOP light is on.

3. Al trips and the KDS ON light is out, and the STDBY RDY light is out.

4-6 Fault Transfer to Standby From KDS

A. Initial conditions include:

Lockout switch in 'NORMAL',
Running in KDS mode

B. One of the KDS faults (listed in section 7-6E1) is detected - DKDS FAULT light is on (as are STOP, KDS-ON, STDBY RDY).

1. MR drops out, standby blowers turn on and 'XFER' picks up.

2. The % REQ relay nearest the operating speed at the time of the fault picks up and in turn closes the associated starting and standby contactors (A2, M2, M3), depending on the speed at time of transfer. STDBY RDY light is out, SPD REQ light is out.

3. Al trips; XFER and IRUN drop out.

4. The drive adjusts to the requested speed.

4-7 Manual Transfer From KDS To Standby

A. While running on the HDS with the STDBY RDY, KDS ON, and STDBY BLWR ON lights on and the lockout switch in 'NORMAL', depress any one of the SPD REQ pushbuttons.

1. Req SPD and XFER relays pick up and requested spd light is on.

2. A2 closes, Al trips, STDBY RDY light is out, KDS ON light is out.

3. M2 and/or M3 close as needed (See 4-2C-4).

4. IRUN, XFER drop out, STDBY LITE is on and drive adjusts to requested speed. When the requested speed is reached, the REQ SPD light is on.

4-8 Test Mode Starting Sequencing

A. Initial conditions include the same as described in Section 4-2A for remote starting sequencing. The Drive Test Module (DTM) should be connected to M01 of the Drive.

B. Before turn-on of the DTM, make sure all logic inputs are open. Follow the instructions on turn-on of the DTM.

C. Turn the DIRECT LOGIC TEST LOGIC INPUT ENABLE switch to ON.

*D. Close LOGIC INPUT SWITCH 1 (local select switch), open LOGIC INPUT SWITCH 2 (not stop), open LOGIC INPUT SWITCH 4 (not KDS request) and close LOGIC INPUT SWITCH 3 (test 50% speed request).

*The close position of the LOGIC INPUT SWITCH is to switch to logic 0.

E. The drive should start and speed should remain on at 50%.

F. Close LOGIC INPUT SWITCH 4 (KDS request). Make sure the lockout switch is at NORMAL position. The drive should transfer from AC Resistor Standby mode to KDS mode.

G. Input the speed reference to SA3 of the DTM for test mode operation (0-15 volt DC).

4-9 Stop From Test Mode

Close LOGIC INPUT SWITCH 2 (Stop). The drive will stop in the manner as described in Section 4-5.

4-10 Bus Transfer During Operation in KDS Mode

The control scheme uses the capability of the thyristor inverters to reduce the rotor-circuit current to zero in less than two cycles during the same time the KDS feeder circuit breaker is being tripped directly by the customer.

1. Upon receiving signals that a bus transfer (fast or residual voltage) is being initiated MR (current limiting resistor contactor) open, suicides firing circuit and current regulator which reduces the rotor-circuit current to zero.

2. At the same time the KDS is receiving the bus transfer signal, the customer signals to open the main breaker M.

3. KDS motor torque will be zero when the current is zero.

4. When the bus transfer signal indicates the completion of the transfer, KDS will signal to close the main breaker M.

5. MR closes, unsuicides the firing circuit and current regulator. The thyristor firing angle returns to normal to restore motor torque to the pre-transfer level or a new level if required by the boiler control.

NOTE: During the residual voltage transfer, an uninterruptable power supply will provide power required for logic and regulating circuits.

4-11 Bus Transfer During Operation In Standby AC Resistor Mode

During operation of the KDS in the standby AC resistor mode, the thyristor inverters are not connected to the rotor-circuit, and therefore are not available to reduce the rotor current to zero during a bus transfer operation. When the KDS main breaker is reclosed at the completion of bus transfer, the standby AC resistors will limit the inrush currents and, therefore the transient torques. Then the drive will be back on line, in pre-transfer operation condition, as soon as the KDS main breaker recloses.

CHAPTER 5
INSTALLATION GUIDELINES

5-l General Requirements for Receiving, Handling and Storing Equipment

A. Receiving Equipment

All items of equipment, before leaving the factory, are carefully inspected and packed by personnel experienced in the proper handling and packing of electrical equipment. Upon receipt of any apparatus, an immediate inspection should be made for any damage sustained while enroute. If damage is evident or an indication of rough handling is visible, a claim for damage should be filed with the carrier at once. Both the transportation company and General Electric Company should be notified promptly. Information as to damaged parts, part number, case number, requisition number, etc., should accompany the claim.

It is recommended that the equipment be placed as close as possible to its permanent location, or at its storage location, before unpacking. Unpacking should be done with care to prevent parts being damaged or marred.

Some parts, such as bolts, screws, etc., are packed in special containers to keep them together until it is ascertained that all parts have been removed from the packing material and quantities checked.

Indoor equipment should be placed under adequate cover immediately upon receipt because packing coverings are not suitable for out-of-doors or unprotected storage.

The equipment may be stored in the same environment in which it was designed to operate. Equipment built for operation in an outdoor environment may be stored outdoors and equipment built for operation in an indoor environment must be stored indoors. (Refer to storage section for additional storage preparation.)

Any items removed from the equipment for storage preparation (or any other reason) must be replaced prior to placing the equipment in service.

B. Handling

Equipment is generally shipped by truck. Small parts or modules may be shipped by air. Vibrations and shock inherent in rail shipment make it the least preferred shipping method.

Refer to equipment outline drawings for preferred methods of lifting and moving the equipment.

The equipment's center-of-gravity should always be carefully observed so that the equipment will not fall.

The limit of shock to which any equipment should be subjected must not exceed an instantaneous force of 3 times the equipment's weight (3 g's for 11 milliseconds maximum).

The limit of vibration to which any equipment should be subjected must not exceed 1 times the equipment's weight (1 g.) at a frequency rate of 10-120 Hz.

C. Only authorized personnel should handle and prepare the equipment for storage. All equipment should be secured against entry by unauthorized personnel.

If any part of the equipment is not to be installed immediately, it must be stored in a clean, dry place and protected from sudden changes in temperature and humidity. If the temperature of the storage room varies to such an extent that equipment surfaces are exposed to sweating or freezing conditions, it must be protected by a safe, reliable heating/cooling system to minimize corrosion by gaseous contaminants (activated by moisture). The internal temperature of the equipment should be kept 5°C to 10°C above that of the storage room or outside temperature (ambient temperature) to eliminate condensation inside the equipment.

It is recommended that for maximum electronic component life that the internal temperature be maintained between +5°C and +45°C. In no instance should the internal temperature of equipment exceed -40°C and +55°C (the maximum temperature of equipment containing CO_2 bottles is +60°C).

Equipment may be stored in environments of 5% to 95% relative humidity with the temperature conditions as stated above to prevent condensation.

To minimize the accumulation of particulate contaminants, a breathable type cover (canvas, etc.) should be used to cover the equipment while in storage.

DO NOT USE plastic type covers (polyethylene, etc.) without specific provisions to prevent condensation beneath this type cover.

For outdoor equipment storage, an inspection should be made to insure that all container openings are sealed and closed against wind, rain, snow, sleet, flooding, dust, etc.

Heaters, air conditioners and/or air purging systems may be required to meet the above storage criteria. If so, it will be necessary to install and check this equipment prior to its operation.

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* *
* CAUTION *
* *
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REMOVE ALL CARTONS AND OTHER MISCELLANEOUS MATERIAL PACKED INSIDE UNITS BEFORE USING HEATERS TO AVOID A FIRE HAZARD.

If equipment has been exposed to moisture, it should be tested and checked by qualified personnel of General Electric Company before being placed into service.

Equipment such as batteries, air conditioners, purge systems, etc. must be inspected and serviced as required by the vendor. Batteries must be recharged every 90 days. The control should be inspected monthly for damage or signs of deterioration and corrective action taken where necessary.

Important parts such as jack screws, gears and chains of circuit breaker mechanisms, linkages and moving machine-finished parts must be prepared for storage as specified in each individual device instruction book.

Low voltage AC drawout breakers (up to 600V) remain in the equipment for storage. Low voltage DC drawout breakers (AG-7 type) may be prepared for storage separately. Medium voltage drawout breakers (5KV to 15KV) should also be prepared for storage separately. Refer to the appropriate circuit breaker instruction book.

D. Special Requirements

There are some cases where special preparation and packaging are required for storage and operation, e.g. - tropical or polar climates. These situations should be referred to General Electric Company for their evaluation and recommendation.

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CHAPTER 6
INITIAL START-UP AND CHECK-OUT PROCEDURES

6-1 Objective

The objective of this chapter is to serve as a sequential guide for inspecting and testing the SILCOMATIC® Plus power converter.

6-2 Precautions

The following precautionary advice is included to forewarn the tester against procedures which might nullify important protective functions designed into the system.

A. Do not, under any circumstances, remove gating signals from only a portion of the bridges paralleled to share the load. The converter is designed to withstand all fault conditions, provided the full complement of bridges is gated and all protective functions are operable.

B. Do not megger electronic control modules and circuits in which solid state electronic devices are present. They can be easily damaged. When meggering converter power circuits, short all buses together to avoid inducing damaging potentials across rectifying devices unnecessarily. Hi-potting the rectifier circuits is not advisable or necessary.

C. Use extreme care and caution when working with or around the rectifier circuits and modules. They are, by nature, high capacity, low impedance paths, and destructive arcing faults could occur by exercising careless procedures.

D. Do not attempt operation of the converter using open or ineffective AC current transformers.

E. Do not remove or insert printed circuit cards unless signal electronics power is off.

F. Do not apply main power to the equipment unless the equipment frame is grounded.

G. Do not operate radio transmitters within a few feet of an operating SILCOMATIC® Plus power converter particularly if the doors are open. The radio frequency (RF) signals generated by the transmitter could induce noise into the unshielded wires within the converter and create erroneous fault conditions.

6-3 Inspection Applicable to New Power Converter Equipment

A. Inspect for proper assembly and connections, both mechanical and electrical.

B. Check for loose, missing or damaged parts.

C. Perform primary breaker (AC) and converter transformer inspection and tests. Set transformer taps to provide rated input volts to the inverter of 760 volts AC for 4000 volts on the primary.

D. Perform insulation megger tests.

E. Inspect mechanical assembly for compliance with assembly diagrams.

F. Check devices having moving parts for freedom of operation, adequate clearance and proper support.

G. Check wire and cables for proper sizing and insulation (especially grounding).

H. Open front doors and remove rear panels and remove any dust, trash, loose parts or any foreign matter.

J. Clear and check fans for easy rotation (without power).

6-4 Test Equipment and Tools

A. A 2-channel oscilloscope with 3 probes (for channel 1, 2 and the external trigger) and 1 current probe. The current and trigger probes must be long enough to allow current to be checked on all cells.

B. Multimeter (preferably digital).

C. Drive Test Module (DTM) or Regulator Test Module (RTM) if available. Be familiar with the instructions for the module being used.

D. Strip chart recorder.

E. Adjustable (plus and minus) DC voltage source (0 to 14VDC) if the DTM or RTM is not available.

F. Elementaries - system: 228B3432, module: 228B3431.

6-5 Test Connections

Connect DTM cabling (if applicable) to receptacle JL located at top of the signal electronics module.

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6-6 TestMode Tests

****
* CAUTION *
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BRIDGE POWER IS NOT REQUIRED AND SHOULD NOT BE APPLIED FOR THE FOLLOWING TESTS. WHEN THE XTFA CARD IS IN THE TEST MODE (PARAGRAPH 6-6 c1), THE BRIDGE POWER IS NOT SYNCHRONIZED WITH THE FIRING CIRCUIT AND EXCESSIVE FAULT CURRENTS CAN BE PRODUCED.

Check the wiring of the following circuits and repair any opens or shorts:

1. The overtemperature (and possible low-oil pressure) switch circuit to the transformer and reactor.

2. The remote reference circuit.

A. Auxiliary Control Power

1. Insure phases 1, 2, and 3 in the incoming side of the control power switch at motor control center (MCC) (KDS Blowers and UPS, unit B4) are connected in the proper sequence.

2. Insure rear panels are in place and doors are shut on SCR panels.

3. Close breaker at unit B4 of MCC, close CPS1-KDS and CPS2-STDBY.

4. Depress PB5 (KDS BLW START) at the remote operator station. Insure each blower at the thyristor bridge is operating properly.

5. Insure transformer T1 is supplying 460vAC. Insure transformers Tl-UPS, T2, CVT1 are supplying 115 $\pm$ 10 VAC.

B. Electronic Power Supplies

1. Determine if power supply PS5 is installed. (See elementary sheet AH or data plate on power supply.) Turn on the Drive Test Module (DTM) which is plugged in M01, and the power supply. Measure the following voltages on the buses at the top of the signal electronic module (or on the DTM meter) for the settings recommended below.

BUS	DTM TEST POINT SELECTOR	RECOMMENDED SETTING	POWER SUPPLY POT
P5	Group B, P5	+5 ±.1VDC	V1
P15	Group A, P15	+15 + .1VDC	V3
N15	Group B, N15	-15 ± .1VDC	V4
choose one of the following:			
N/A	Group A, P28/2	+14 ± .7VDC	V2
P28	N/A	+28 ± 1.5VDC	V2

C. Test Oscillator and Gate Pulse Generation

1. Insure CPS1 and CPS2 are off. On both XTFA cards, which are mounted on the control panel rear wall above the signal electronic module (M02), disconnect the ribbon connector from JE and reconnect it to plug JF. This results in the test mode configuration. Turn CPS1 and CPS2 on. verify the LEDS on the cards (both sets) indicated below.

LED	CARD	STATUS
TST	on NFOD	on
SEQ	on NFOA	off
PLL	on NFCC	off

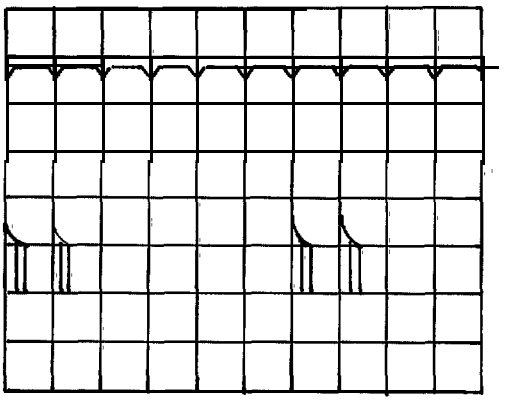
Note: It may be necessary to reset the NFOB cards during this test.

2. Do the following for one set of cards at a time, Connect a test reference to the DTM SIGNAL ACCESS input SA1 (A-Bridge) (Or to testpoint TREG on the NFCC card). use SA2 for B-Bridge.

3. Connect channel 1 of the scope to COS (test point 9 on the NFCC card). Use the external scope synch connected to TP-PH2 (NFOB card). Monitor the gate current of cell 1 on the A (B) inverter bridge (inside A-Bridge [B-Bridge] SCR door) with the current probe on a second channel. The current probe is connected to the white wire for the following amplitude and position checks.

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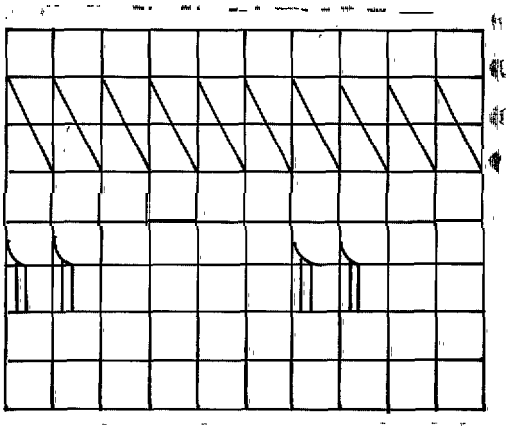
4. Release the firing circuit suicide with the DTM by moving rotary switch to release FC (or connect pin 64 NSCA to DCOM), set the test reference output for -9.0 volts. The oscilloscope presentation should appear as follows:



Horizontal: 60°/DIV
Channel 1 : 2V/DIV
Channel 2 : 1.0 A/DIV

5. Decrease the test reference- output to 0.0 volts and verify the COS waveform appearing on channel 1 of the scope retards smoothly.

6. Set the test reference output to 12.0 volts to obtain the maximum retard. The scope presentation should appear as follows:



Horizontal: 60°/DIV
channel 1 : 2V/DIV
Channel 2 : 1.0A/DIV

7. Check the remaining forward cells in the master bridge for proper firing order and presence of pulses. As the firing sequence is followed (1,2,3,. . . 6), the firing pulses will lag the previous cells firing pulses by 60° (1 grid on 2MS setting).

8. Repeat 2-7 for the B-Bridge set of cards.

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D. Preliminary Set-Up Checks

1. Operation from KDS

a. Rack the main breaker (M) into TEST position.

b. Connect a signal generator to TBCL AA05P and AA06P (or a test voltage source to pin 08-B1K), to simulate the speed feedback of the motor.

c. Turn on control power (CPS1, CPS2 and CPS5).

E. Fault Signal Checks

1. The following KDS faults will cause the drive to transfer to the standby mode - provided the lockout switch is in "NORMAL".

	FAULT	INDICATION
a.	Shorted diode (with drive running).	HLEA - 2G CR5 on
b.	10% difference in speed feedback and rotor volts	HLEA - 2H CR4 on
c.	2nd suppression	NFCC - SUP on
d.	Blown control fuse	HSCD - LED F off
e.	Thermal overload (A or B inverter)	NFOC - TOL on
f.	Reactor overtemperature (A or B channel)	HLEA - 2G CR5 on
g.	DC overvoltage (A or B channel).	NFCC - OV on
h.	Transformer overtemperature	HLEA - 2F CR6 on
1.	KDS blower loss	HSCD - LED E off
j.	Switch 1, 2, 3 or 4 opens	HSCD - LED D off
k.	Overvoltage	NFCC - OV on

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2. The following faults will trip MR (running contactor) and reclose once, if faults persists (reoccurs on reclosure) drive will shut down.

   a. Suppression (instantaneous inverter overcurrent set at 1.4 times current limit or 210% full load).

   b. Inverter fault detector (failure of inverter to commutate).

3. The following faults cause the drive to shut down, not transferring to standby - but trip the main breaker (M) directly and consequently the other contactors.

   a. Switchgear faults - which include motor overcurrent, motor current differential, transformer overcurrent.

   b. AC fault - logic overcurrent persisting for more than 25 msec, in A or B inverter bridge.

   c. Loss of standby blower while in standby mode or when attempt to transfer to Standby from KDS and standby blowers don't start.

   d. Stall - During starting in AC resistor mode, if the motor does not turn after the main breaker (M) and A2 contactors had closed for 2 seconds the main breaker (M) and all contactors will trip. Reset at the remote operation station before restart.

4. With no faults and with KDS lockout switch in "NORMAL", FAULT light will be out. Simulate each of the conditions below as follows. Verify that FAULT light comes on for each test. Most faults must be reset by the SRST on the NFCC card.

   a. Shorted diode. Connect a positive test voltage to TP7-1B (elementary, sheet 310Q) and adjust to obtain 2V on (TP8-1B) M01. Comparator will pickup and CPA-1E MO1 will light. FAULT light will light.

   b. Speed/Current Differential. Connect a positive test voltage to TP5-1B (elem. Sheet 310N) and adjust to obtain 1V on (TP6-1B). Comparator CPC-1D will pick up. FAULT light will light.

   c. Blown Control Fuse. Temporarily remove fuse FU1 (elem. Sh. AB). FAULT light will light.

   d. Thermal Overload. Connect a test reference to T1A (TP3) (elem. Sh. 3206) A-inverter bridge. Raise to obtain 11 volts on CFB. ILR will light after a time delay. Repeat for B inverter bridge (elem. sh. 320K).

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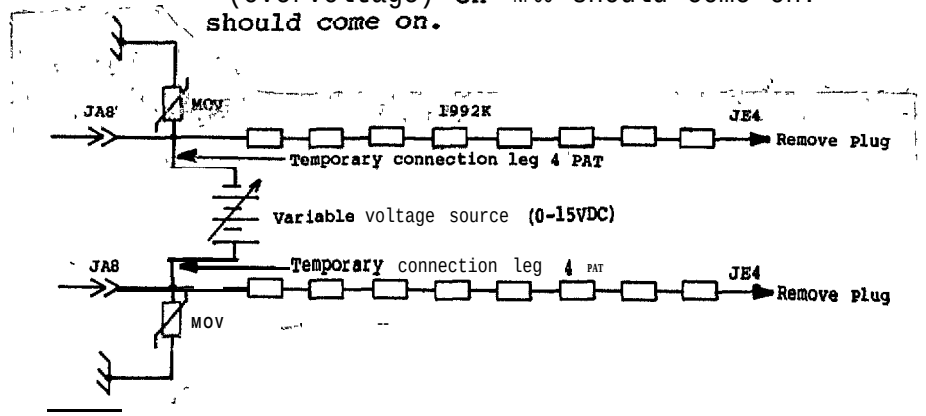
e. Reactor Overtemperature. Disconnect the lead from one of the reactors. FAULT light should light. Replace lead and repeat for other reactor.

f. DC Overvoltage. Connect a voltage source to PATT and NATT on XTFA board (elem. sh. 330A). Raise voltage to obtain 9V on VFB-NFCC (elem. sh. 320H). OV (overvoltage) on NFCC should light, FAULT light should light. Repeat for B inverter bridge (elem. sh. 320c and 320L).

g. Transformer Overtemperature. Temporarily remove lead from the transformer overtemperature device at inverter transformer. FAULT light should light.

h. KDS Blower Loss. Temporarily trip each blower in Kramerstat. FAULT light should light.

i. Overvoltage. Turn off bridge power and control power. Remove plug JE from master PAT A-Conv, leg 4 (removes NEG bus feedback). Remove plug JE from master PAT A-Conv, leg 5 (removes POS bus feedback). Connect a temporary wire to ATT on both master PATS as shown below. Connect a voltage source to these connections (0-15VDC) positive on leg 5, negative on leg 4. Turn on control power only. Raise the voltage source to 10.6 volts. VFB on NFCC reads 8.86V. LED OV (overvoltage) on NFCC should come on. FAULT light on door should come on.



5. Suppression (Neglect the AC Fault Indication)

Temporarily remove R60 and R61 from daughter board Of XTFA (elem. sh. 330A and 330C). Connect a voltage source to IA test point (NFCC) (elem. sh. 320G and 320K). Input a step voltage of 10.80 volts and immediately remove it. SUP light (NFCC) should come on, FAULT light should remain off, MR should open but M should remain closed. After 1.5 seconds, the SUP light should go out and MR should reclose. Repeat but leave the 10.80 volts on. The sequence should repeat except that after MR recloses, FAULT light and SUP light should both come on and all contactors should trip. Repeat for B-bridge (elem. sh. 320K).

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6. with no faults and lockout switch in standby, simulate each of the following faults. Main switchgear contactor (M) will drop out.

   a. Switchgear Fault. Temporarily activate lockout relay at switchgear. FAULT light should light, contactors should trip.

   b. AC Fault. Temporarily remove daughter boards (DTFA) from XTFA for both A and B bridge (elem. sh. 330A and 330C). Connect a test reference to IA-A bridge (elem. sh. 320G) and slowly raise to 9.5V. ACOC on NFOD card location B1E should light, the fault light should light and all the contactors should trip. Repeat for IA-B inverter bridge (elem. sh. 320K).

   c. Loss of Standby Blower. Depress STANDBY BLOWER STOP PB (standby blowers should stop) initiate a KDS transfer fault by jumpering pin 34-A2F to DCOM. Fault light should light, contactors should trip.

F. Thermal Model verification

NOTE

Voltages applicable to this test appear on elementary sheet 320Q and 320R.

1. Set PS5 off.

2. Connect a voltmeter (preferably digital) and a channel from the chart recorder to -TJ (testpoint 6) on the NFOD card.

3. Set PS5 on.

4. Connect a test reference to A-bridge to test point TIA on the NFCC card.

5. Release the firing circuit suicide.

6. Adjust the test reference applied to TIA, NFCC board, so CFB (FWD) will be applied as a stepped voltage (switch on, then switch off).

7. Allow test point -TJ to settle to its ambient value (TAMB).

NOTE

This step may result in an excessive waiting period, however, the -TJ ambient value may be expedited by momentarily jumpering pin 13 of the NFOC card to either N15V or P15V on the back of the signal electronics module.

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8. When test point -TJ value reaches TAMB, start the chart recorder, apply the test reference output as a step and observe the chart recording. At TF seconds, the chart recorder should indicate a change of voltage equal to ΔTHFM; if not, refer to the thermal model calibration procedures.

9. This completes the thermal model verification.

10. Repeat for B-bridge (elem. sh. 320R).

G. Sequencing Checks

Use REF 1 of DTM input reference to **TP3-1G M01.**

1.0 Standby Sequencing (Remote)

1.1 KRAM LO SW in NORMAL, Standby blower on test switch closed, lockout relay to test switch closed, logic enable SW in DTM off. No AC fault KRAM no fault KRAM - Adjust 50% timer pick up to 117 seconds, 69% timer to 35 seconds, 79% timer to 13 seconds, Al will remain open during this test.

1.2 Repeat the STANDBY SEQUENCING test with KRAM L.O. switch in KRAM NO CHANGE.

ACTION	A2	LED4-2A M01 (M)	REMOTE LIGHTS X = ON																			X=Close	
			Standby Ready	50% Req	69% Req	79% Req	50% Spd	69% Spd	79% Spd	KDS Ready	KDS On	KDS Fault	STOP	KDS Req	KDS BLW Off	STDY BLW Off	KDS BLW On	STDY BLW On	M2	M3		
Initial Condition	**Open**	**Off**	X									X		X			X					
Depress & release PB50	(Closed	**On** after IA2 **closed**		X			117S X								X			X				
Depress & release **PB69**	NC	NC			X			35S X							X			X	x			
Depress & release PB79	NC	NC				X			13S X						X			X	x	X		
Depress & release PB69	**NC**	NC			X			X							X			X	X			
(Depress & release PB50	NC	NC		X			X							X				X	X			
Depress & release PBSTOP	**Open After M** **open**		X										X		X			X				

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2.0 Transfer From Standby to KDS (Remote)

2.1 Set up same as 1.1, connect signal generator per 6-6-D-C. Set at 0.0 Hz. Start drive on STANDBY at 50% speed.

ACTION	A2	A1	MR	LED4-2A M01 (M)	REMOTE LIGHTS X = ON																X=Close		
					Standby Ready	50% Req	69% Req	79% Req	50% Spd	69% Spd	79% Spd	Sax Ready	Sax On	KDS FAULT	Pollé	KDS Req	Off	330 STDBY BLW	KDS BLW On	STBY BLW On	M2	M3
	X=Closed			X=On																			
Initial Condition	X			X		X			X								X			X			
Depress & release PB5 (KDS BLW)	X			X		X			X	X							X		X	X			
Throw L.O. SW to KRAM depress & release PB1	X			X		X			X	X									X	X			
Return L.O SW to KRAM depress & release PB1	X		X	X		X			X							X			X	X			
Adjust Sig. Gen. to 720 Hz	X	X	X	X		X			X		X					X			X	X			
Apply 1.0 volt to TP14-1G M01		X	X	X	X								X						X	X			
Jumper AN13H to AN08K, depress & release STOP		X		X												X				X	X		
Remove the Jumper AN13H to AN08K		*			X						–				X	X				X			

* Open after M open

2.2 Repeat above with drive started on standby at ≤9% speed.

2.3 Repeat above with drive started on standby at 79% speed.

3.0 Manual Transfer From KDS to Standby

3.1 Start drive on standby at 69% speed and transfer to Kramerstat per 2.1. Do not depress STOP.

ACTION	A2	A1	MR	LED4-2A M01 (M)	REMOTE LIGHTS X = ON																		X=Close	
	X=Closed			X=On	Standby Ready	50% Req	69% Req	79% Req	50% Spd	69% Spd	79% Spd	KDS Ready	KDS On	KDS FAULT	STOP	KDS Req	KDS BLW Off	STDBY BLW Off	KDS BLW On	STBY BLW On	M2	M3
Initial Condition	X	X		X	X								IX					IX	IX			
Throw L.O. SW to STANDBY	IX	IX		X									IX					IX	IX			
Open STANDBY BLW on SW. Return L.O. (SW to NORM	IX	IX		X									IX				IX	IX				
Depress PB50	IX	IX		X									IX				Ix	IX				
Close STANDBY BLW ion SW. Depress PB50	X			X	IX	IX			X*			IX						IX	Ix			

* After 117S

3.2 Repeat above with signal generator adjusted to 60% speed. The same except 69% Req and 69% Spd lights come on and M2 closes in final step (regardless of which request pushbutton is depressed).

3.3 Repeat above with signal generator adjusted to 72% speed the same except 79% Req and 69% Spd lights come on M2 and M3 closed in final step.

4.0 Fault Transfer to Standby

4.1 Start drive in STANDBY at 50% spd and transfer to KDS per 2.1. Do not depress STOP.

ACTION	A2	A1	MR	LED4-2A M01 (M)	REMOTE LIGHTS X = ON																				X=Close	
	X=Closed			X=On	Standby Ready	50% Req	69% Req	79% Req	50% Spd	69% Spd	79% Spd	KDS Ready	KDS On	KDS FAUL T	STOP	KDS Req	KDS BLW Off	STDBY BLW Off	KDS BLW On	STBY BLW On	M2	M3				
Initial Condition	X	X		x	X								X					X	X						
Depress PB6 (KDS BLW STOP)	X	I	I	X		X			X					X						X					

4.2 Repeat above with signal generator adjusted to 60% speed. The same as above except 69% Req and 69% Spd lights come on, 50% Req, 50% Spd off.

4.3 Repeat above with signal generator adjusted to 72% speed. The same as above except 79% Req and 79% Spd lights come on, 50% Req, 50% spd off.

5.0 Local Sequencing

5.1 KRAM L.O. SW in NORM, lockout relay test switch closed, STANDBY BLW test switch closed, logic enable switch in DTM ON, logic switches 1, 2, 3, 4 open, no KRAM fault - depress PE5 (RDS BLW START) connect SA3 to REF 1 of DTM, set at zero.

ACTION	A2	A1	MR	LED4-2A M01 (M)	REMOTE LIGHTS X = ON																	X=Close	
					Standby Ready	50% Req	69% Req	79% Req	50% Spd	69% Spd	79% Spd	KDS Ready	KDS On	KDS FAULT	STOP	KDS Req	KDS BLW Off	STDBY BLW Off	KDS BLW On	STBY BLW On	M2	M3
	X=Closed			X=On																			
Initial Condition I					X							X			X				X	X		
Close & Open DTM Logic SW 3 (50% REQ)	X			X		X			X*			X							X	X		
Close 6 Open DTM Logic sw 4 (TKDS REQ)	X		X	X		X			X			X				X			X	X		
Apply 1.0 Volt to I TP14-1G	X		X	X	X								X						X	X		
Closed & Open DTM Logic SW 1 (Local Select)		X	X	X	X								X						X	X		
Closed & Open DTM Logic SW 2 (TEST STOP)					X							X		X				X		X		

• 12.5 Seconds

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6.0 Bus Transfer Check

6.1 Temporarily connect two test switches to simulate the Bus Transfer signal from customer (AP32K and AC1, P105 and AR10E).

6.2 Open test switches.

6.3 Simulate the motor is running at 50% speed (see 5.0 above) at KDS mode.

6.4 Close test switches, MR should open, firing circuit and current regulator should suicide.

6.5 Open test switches, relay MXBX (AN13) should close for 2 seconds and drop out, MR close, firing circuit and current regulator unsuicide, main breaker reclose if customer's other interlocks are satisfied.

6.6 Remove test switches; end of sequencing checkout.

6.7 Repeat bus transfer checks using customer's initiating signals and interlocks to verify proper operation of external control scheme.

6.8 End of sequencing checkout and test mode.

6-7 Power Mode Checks

A. Phasing Check

1. On both XTFA cards disconnect the ribbon connector from plug JF (test mode) and connect to JF for normal mode.

2. Remove jumpers from AA25B 27B, AN71E and AM76 and reconnect external connections. Also remove the N.O. MX contact at AN08L and AN08K.

3. Remove main power connections from the rotor of the motor (these leads will have 1260V on them during this test and must be clear of each other and ground). May be done by opening Al and A2 contactors.

4. Set DTM logic SW 1 (local select) to 0.

5. Turn on STANDBY fans and KDS fans.

6. Remove plug JC from XTFA cards (this will prevent cells from firing).

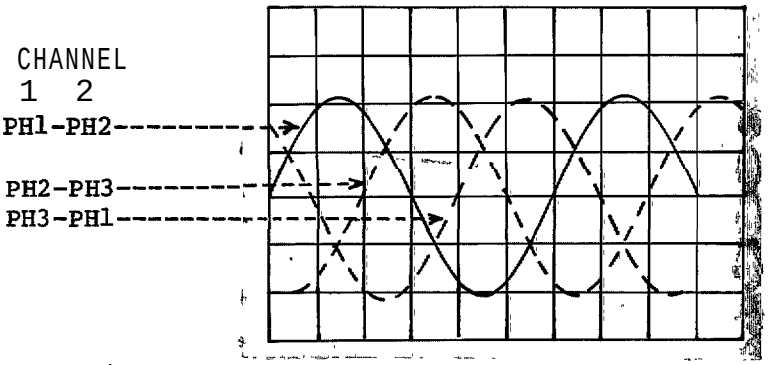
7. Connect a test reference to TP1-1C (elem. sh. 310Y). Adjust for 0 volts.

Verify the following LED status:

LED	LOCATION	STATUS
TST (red)	NFOB-BlC + B2C	Off
SEQ (red)	NFOD-BlE + B2E	Off
PLL (red)	NFCC-BlA + BlB	Off

All cell voltage neon lamps of both bridges should be on.

9. Set the scope to read differentially. Connect external trigger to PH2 test point NFOB-BlC. Connect scope leads as indicated below and observe the wave forms shown below. Repeat for B-bridge NFOB-BZC.



10. Observe PB-NB test points (NFOB-BlC) differentially. The wave form should remain centered at the 0 volt ref line." Repeat for B-bridge.

11. Set DTM logic SW 2 to 0 (NFOB-B2C). Enables test stop. Main breaker should trip.

B. Load Test and Current Regulator Response as a Rectifier

1. With control power off, install plug JC on the XTFA cards. All other connects 6.6A 1 to 8 remain as they were. Remove the signal diode on slot G3 of NGPA board 1F to allow the inverter to act as rectifier.

2. With lockout mode switch in 'NORMAL' and with DTM in unsuicide FC, set the DTM logic SW 2 (test stop) to 1.

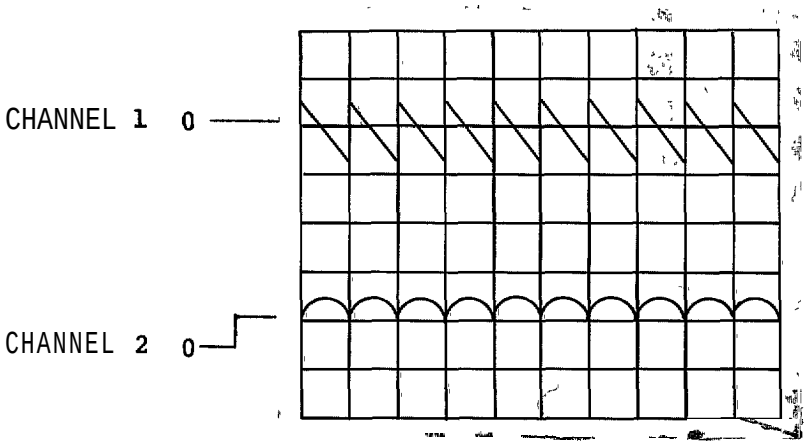
   a. Main breaker closes.

   b. Red LEDs on PATs should be lit indicating pulses are present at the cells.

   c. Firing angle should be 90° retarded, check both bridges.

d. PB-NB observed differentially should indicate zero volts. (Check both bridges).

3. Observe test point CFB (NFCC-BlA) on channel 2 and VFB (NFCC-BlA) on channel 1. Trigger scope from PH2 NFOB. Gradually increase reference at TPl-lC (current ref) until CFB (NFCC-BlA) has about 1 volt. Wave shape should be as follows: (PB-NB no longer 0).



HORIZONTAL: 60°/DIV
CHANNEL 1: 4V/DIV
CHANNEL 2: 2V/DIV

Repeat for B-bridge (NFCC-B2A).

4. Move DTM to suicide all - current and voltage should go to zero (both bridges). Depress stop button. Main breaker trips.

5. Connect two channels of a Brush recorder to CFB and test Kef input TPl-lC and follow 6.7B 1 to 4.

6. Increase the steady reference by a small step. The response of the current should be about 300 radians. (.0033 sec) A-bridge.

7. Repeat 6 for B bridge.

8. Suicide by moving DTM to suicide all and set DTM logic SW 2 to 0.

9. Replace the signal diode in C3 of NGPA board 1F.

6-8 Power Checks as A KDS and Speed Regulator Response

1. Reconnect the motor stator cables.

2. Remove Brush leads and test reference into current regulator.

3. With DTM in release firing circuit mode:

(a) Connect a test reference into SAl (A-Bridge) and SA2 (B-Bridge) DTM (or TREG-NFCC-1A + 1B).

(b) Connect a DC voltmeter between TP15-A1E (elem. sh. 310T) and ACOM.

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(a) Connect a test reference into SA1 (A-Bridge) and SA2 (B-Bridge) DTM (or TREG-NFCC-1A + 1B).

(b) Connect a DC voltmeter between TP15-A1E (elem. sh. 310T) and ACOM.

4. With lockout mode switch in 'NORMAL', toggle T50% REQ SW (DTM SW #3) to l-0-1 (off-on-off) and before motor accelerates to 50% speed, set test stop SW (DTM SW #2) to 0 (on). Observe speed feedback for proper polarity and approximately correct value.

5. Speed Regulator Response

A. Put lockout mode switch in 'NORMAL' and turn STANDBY blowers and KDS blowers on. Toggle T50% REQ SW (DTM-SW #3). Toggle TKRAM REQ SW (DTM-SW #4). Set LOCAL SW (DTM-SW #1) on (local mode).

B. Connect a Brush recorder to DTM-GB2 (elem. sh. 310T) (spd fdbk) and to TP2-A1K (NSCA card) (elem. sh. 310X).

C. Apply a small step input to TP2-A1K NCSA card (to increase speed). The response of the SPD FDBK should be about 2.5 Radians. (.04 sec).

6. Move the DTM to the off position. Run the drive as described in sequencing checks 6-9 1, 2 and 3.

This completes the power checks.

6-9 Sequencing Check in Power Mode

1. Notes On Indication

A. STOP light is on any time the incoming main breaker (M) is open - i.e., any time the motor (or transformer) is deenergized.

B. Any SPD % REQ PB that is depressed should light indicating it was requested provided the STANDBY READY light is on. Any other % REQ PB's lights should be off. Also SPD lights indicate the speed the motor is running at whenever the drive is running on the AC resistors. (These are not direct reading indication).

C. The firing circuit (FC) and consequently the speed regulator (SR) are suicided unless in the KDS mode. When FC is unsuicided, a 90% retard ref is applied and a SR preconditioning is applied. The KDS accepts current and when 20% FL amps are detected, the SR preconditioning is released and follows the ref, and the 90° retard is removed. Now the KDS is officially on and is under operator control.

D. The BLOWER START PBS (STANDBY and KDS) light whenever the blowers are running.

E. The KDS REQ light is on only long enough for KDS to draw current, then the KDS ON light is on instead.

2. Sequencing Check (Standby Mode)

A. Lockout switch in 'NORMAL'.
Depress and release (D + R) start STANDBY BLOWERS PB - STANDBY READY and STOP lights on, all contactors remain open, FC is suicided. Stop STANDBY BLOWER lights from ON to OFF.

B1. D + R 50% request (50% REQ relay picks up).
Contactors: M and A2 close. Lights on - 50% REQ. Lights off - STOP, STANDBY READY, DTM GB2 = -5V, FC is suicided, 50% TIMER times out. Lights on - 50% running after 50% timer picks up.

B2. D + R 69% Request (69% REQ relay picks up, 50% drops out).
Contactors: M2 closes. Lights on - 69% REQ, stop, DTM GB2 = -6.96V, FC suicided, 69% TIMER times out. Lights on - 69% running after 69% timer picks up - Lights off - 50% REQ, 50% running.

B3. D + R 79% REQ (79% REQ relay picks up-69% drops out). Contactor M3 closes. Lights on - 79% REQ. DTM GB2 = -7.9V, FC suicided, 79% TIMER times out. Lights on - 79% running after 79% timer picks up. Lights off - 69% REQ and 69% running.

B4. D+R 69% REQ. Contactor M3 trips. Light on - 69% REQ. and 69% running. DTM GB2 = -6.96V, FC suicided. Lights off - 79% REQ and 79% running.

B5. D+R 50% REQ. Contactor M2 trips. Lights on - 50% REQ, and 50% running. DTM GB2 = -5V, FC suicided. Light off - 69% running.

C. D+R STOP PB.
Contactors: M, then A2 trip. Lights on - STOP, STANDBY READY. DTM GB2 = OV. Light off 50% running and 50% REQ.

D. D+R 79% REQ.
Contactor M, A2 close. After 117 seconds, M2 closes, DTM GB2 = -5v. After 35 seconds, M3 closes, DTM GB2 = -6.96V.
Lights on - 79% REQ and 79% running. DTM GB2 = -7.9V, FC suicided. Lights off - STOP, STANDBY READY.

E. D+R Stop STANDBY blowers.
Contactors M, then A2, M2, M3 trip. Lights on - STOP and stop STANDBY BLOWER. DTM GB2 = 0.

F. Repeat A through E but with lockout switch in 'KDS' position.

3. Sequencing Check (Transfer from Standby Mode to KDS Mode)

A. Lockout switch in 'KDS' position. D+R Start STANDBY blower, start Kramerstat blowers, KDS REQ. Contactors closed - none. Lights on - STANDBY READY. Lights off - KDS REQ.

B1. Lockout switch in 'STANDBY'. D+R KDS REQ. Contactors closed - none. Lights on - STOP. FC is suicided, lights off - KDS REQ.

B2. Lockout SW in 'NORMAL'. D+R KDS REQ. Contactors closed - none. Lights on - STOP, Lights off - KDS REQ.

C. D+R 50% REQ. Contactors: A2 closes. Lights on - KDS READY (when DTM GB2 = -5V); 50% REQ. and 50% running after 50% timer picks up.

D. D+R KDS REQ, IRUN picks up, MR closes, FC + SR are unsuicided, 90" retard applied. Contactors: MR and Al close. After KRAM reaches 20% FL current, A2 trips. Lights on - STANDBY READY, KDS RUNNING. DTM GB2 SPD FBK (-5V min - -lOV max) follows speed ref. KDS REQ light on after KDS REQ was depressed and before speed regulator is unsuicided.

E. DCR STOP. Contactors MR, then M, then Al trip. Lights on - STANDBY READY. FC - suicided. Lights off - KDS running, STANDBY running.

F1. D+R 50% REQ. Contactors - M and A2 close. Lights on - 50% Running, KDS READY, 50% REQ. FC - suicided.

F2. D+R KDS REQ. Contactors - MR and Al close, A2 trips. Lights on - STANDBY READY, KDS RUNNING. FC + SR unsuicided and drive follows speed ref. Vary the 4-20 milliampere reference from customer, the speed of the motor should follow.

G1. Move lockout switch to 'STANDBY'. No change except STANDBY READY light - off.

G2. Move lockout switch to 'KDS'. Contactors M, MR, and Al trip. Lights on - STOP. Motor coasts down. Lights off - KDS running.

4. Sequencing Check (Transfer from KDS Mode to Standby Mode)

A. Lockout switch in 'NORMAL', D + R START STANDBY blowers, start KDS BLOWERS, and 50% REQ. Lights on - start STANDBY blowers, start KDS blowers, STANDBY READY, KPS ready, KDS running.

B. Perform step F1 and F2 so that drive is running in KDS mode.

C. Adjust ref so that SPD FDBK < 59% SPD - LED CPC-A1E off.

D. Cause a KDS fault by jumping pin 34-A2F to ACOM. Lights on: KDS FAULT, 50% REQ and 50% running and KDS fault contactors: Al, MR trip; A2 closes (M still closed). Lights off - KDS running.

E. Remove jumper (fault light goes off) t depress KDS REQ PB. Lights on - KDS ON. Contactors: M, Al, MR are closed. Light off - 50% REQ and 50% running.

F. Adjust ref so that SPD FDBK >59% SPD. CPC-A1E on, CPC-A1F off.

G. Repeat D and M2 should also close. Verify transfer to 69% running on STANDBY.

H. Repeat E and adjust SPD FDBK >74% SPD. CPC-A1F on.

I. Repeat G and M3 should also close. Verify transfer to 79% running on STANDBY.

J. Repeat E and move lockout switch to 'STANDBY'.

K. Jump pin 34-A2F to ACOM. All contactors should trip. Lights on - STANDBY READY, and KDS FAULT, STOP. STANDBY blowers running, KDS blowers running, all other lights off.

L. Remove jumper then repeat A, B, C, then cause an AC fault by jumping pin 42-A2F to DCOM. Lights on: STANDBY READY, FAULT, STOP blower lights running. Contactors: all tripped. All other lights off.

5. While in KDS mode lockout switch in 'NORMAL', with standby fans are off, speed <54%, depress 79% req PB. Contactors: Al + MR trip; A2, M2, M3 closed. Lights on: STANDBY FAN, KDS READY, 79% REQ, and 79% SPD, FC suicided.

6. At the completion of above tests, verify that customer's 4-20 milliampere signal gives the correct range of operating speed in the Kraemer mode.

3/4/85 CS

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CHAPTER 7
PREVENTIVE MAINTENANCE

7-1 Inspection Routine

A periodic routine of cleaning and inspection should be established. Such cleaning is necessary to prevent faulty operation of a device due to accumulation of dust and corrosion. Inspections should be more frequent if the demands on the equipment are exacting. under normal conditions, the protective devices do not operate; therefore, these devices need careful inspection. If the equipment is in operation when being inspected and a short interruption of service can be permitted, the manual tripping of a protective device will afford an opportunity to observe the manner in which the equipment is functioning.

If at any time a device fails, the cause of the failure should be located and corrected before any remedy is applied and, if necessary, the part adjusted or replaced.

NOTE: When ordering parts, give a detailed description of the part and complete nameplate data on the device.

For specific maintenance information on the various devices, refer to the particular device instruction book.

****
* *
* CAUTION *
* *
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UNDER NO CIRCUMSTANCES SHOULD AUTOMATICALLY CONTROLLED EQUIPMENT BE STARTED BY HAND UNTIL THE CAUSE OF THE FAILURE TO START AUTOMATICALLY IS DETERMINED AND CORRECTED.

7-2 Records and Log Book

It is recommended that a regular inspection and maintenance program be employed, and that a log book be kept. This procedure will help detect malfunctioning equipment and determine the causes of the trouble. It is suggested that a form be made up and filled out at each regular inspection or when trouble is experienced. The form should contain the following data:

Normal Operation:

Operator Converter DC amps
Date and time Maintenance performed
Converter DC volts Comments

Abnormal Operation - Additional Observations:

Alarm lamps
Tripped breakers
Circumstances at time of trouble, such as:

Starting up
Shutting down
Load
Test operation
Disturbances

Abnormal operations elsewhere but still associated with the converter.

7-3 Troubleshooting - General

A preventive maintenance program will help reduce costly downtime. Should trouble occur, the following tips will help get inoperative equipment back into service with minimum delays.

1. Become familiar with each circuit and the operation of each new component as it is installed.

2. Keep the elementary diagram handy. This is the most valuable tool the manufacturer can furnish for use in troubleshooting.

3. Keep the portable instruments handy for checking voltage, current, resistance, etc.

4. Have on hand adequate spare parts to replace failed components.

After operating properly, should the drive become inoperative, or its performance change, care should be exercised during troubleshooting in order not to get the drive out of adjustment. Generally, the difficulty stems from a loose connection, broken wire, or failed component and can be corrected without making additional adjustment to the control. Considerable time will be saved if the difficulty can be corrected without changing the rheostat settings necessitating retuning the drive. When the system is tuned up initially, pertinent data should be recorded - test point readings, etc. The following sheets give some troubleshooting hints. If a printed circuit board is changed and does not correct a trouble, replace the original card in the panel.

****
* CAUTION *
****

DO NOT INSERT OR REMOVE A CARD WITHOUT FIRST TURNING OFF POWER. FAILURE TO DO SO MAY RESULT IN DAMAGE TO CIRCUITRY.

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I. If the drive is shut down and the reason for the shutdown is unknown, the following checks should be made first.

1. Check the neon cell voltage lights on the HPTF to see if voltage is present across all cells. A light which is out indicates a failed cell or a failed light.

2. Check the SUP LED on the NFCC card. If the LED is on, an overcurrent condition has been applied to the drive. Check the drive to see if there is any obvious mechanical or electrical reason for the fault. Correct any improper operating condition before depressing the reset switch on the NFCC card.

3. If there is no evidence that the shutdown was due to an overcurrent fault, the following points should be checked:

   a. Is the Phase Lock Loop (PLL) LED on? If off, the PLL is operating correctly. If the light is on, make sure that the AC power is on as evidenced by the cell voltage lights. Check that the cable from XTFA to NFCC is correctly and firmly in place. Check that the Status LED (yellow) on power supply is on. If the PLL LED is on and none of the above tests determines the cause, a new NFCC card should be tried.

   b. If the PLL LED is off, check to see if the LED indicators on the HPTF are on. If the LED indicators do not turn on, a new NFCC should be tried. If the LED indicators do turn on, the NFCC is probably operating correctly.

   c. Check the 120 volt AC control power.

   d. check the fuses (FU5, FU8, FU9 and FU10) at the control power.

   e. Check the SCR gate pulse by means of a current probe. Place the drive in test mode by moving the XTFA board JE connector to receptacle JF. The purpose of the test mode is to allow gate pulse generation and converter check-out with no bridge power.

      (1) The RED LED on NFOB card labelled TST should be luminous.

      (2) Arrange the DTM to inject a signal into SA1 (DTM).

      (3) Monitor the gate current of SCR cell with the current probe to an oscilloscope.

      (4) Use the DTM to release the firing circuit suicide.

      (5) Vary the REF at SA1 from -8 to +8, at any ref level, the gate pulse should look the same except the position the magnitude of the last pulse train should at least have 200 milliamperes to start with. If the gate pulse of any gate is not good, change the PAT card and test again.

(6) To place the drive back to normal mode by moving the **XTFA** board JF connector to receptacle JE.

(7) Reset the NFCC card and start the drive.

II. If the drive is shut down and the fault indicating light is on, it means that any one of the conditions for which the **Fault** Signal illuminates causes the fault. Check the LED for "telltale" illumination and correct the error before resetting the NFCC card, and subsequently the drive.

III. When the **IMOK** light (yellow LED) is off in any electronic cards, it indicates that the card is either in a fault condition or that a monitored op-amp output has reached a fault saturation level. The IMOK LED does not latch in the tripped (off) state.

CHAPTER 8
ADJUSTMENTS

8-0 This chapter gives basic procedure for making adjustments to various circuits found in the elementary. No adjustments should be necessary after the drive has been operating properly, even if a care is replaced as the variable components are on a moveable (daughter) board that can be moved to the new card.

Most of the following adjustments are already included in the electronic module elementary 228B3431, but are duplicated here for further clarity.

TWO elementary sheet numbers are listed for the drive; the first is from the system elementary 338B3432, the second (300 series no.) is from the electronics module elementary 228B3431. The electronics module provides more detailed information (backplane pins, secondary testpoints, etc.) on the circuits.

When two functions are equally applicable, the 2nd will be put in brackets [I. (Information applicable to the B-bridge is also included in brackets as the procedure is the same as A-bridge).

8-l Speed Reference

8-2 Speed Feedback

8-3 Current Feedback

8-4 Voltage Feedback

8-5 Speed/Volt Comparison

8-6 20% Current Detector

8-7 Shorted Diode Detector

8-8 Tach Preconditioning

8-9 Cur rent Regulator

8-10 Speed Regulator

8-l Speed Reference (elem. sh. AX and elem. sh. 310U)

A. Remote Process Signal

1. Turn R4-1H fully CCW, R6-1G fully CW, and turn R1-1H fully CW.

2. Apply the minimum process signal (4MA) and turn R4 CW until GB3 (DTM) = 0.0 volts.

3. Apply the maximum process signal (20MA) and turn R6-1G CCW until GB3 (DTM) = +5.0V.

4. Increase the process signal above 20MA. (Add a (voltage) test ref (-) to TP14-1H if unable to increase current ref.). GB3 should increase above +5.0V. Turn R1-1H CCW until GB3 is reduced to +5.0V.

B. Local Manual Ref.

1. Turn R6-1H fully CW.

2. Apply a test ref of -15v (or maximum available) to SA3 (DTM) and turn R6-1H CCW until GB4 (DTM) = +5.0V.

C. Timed rate reference (elem. sh. AY and elem. sh. 310V)

1. Turn RH2-1C fully CW, local/remote SW in remote [local] remote [local] [-15V].

2. Apply the maximum reference (20MA) and verify that GB3 (DTM) = [GB4 (DTM)] +5.0V.

3. Connect a Brush recorder to TP15-1G.

4. Remove the spd ref and allow TP15-1G to go to 0.0V, reapply the reference (as a step input) and adjust RH2-1C so that TP15-1G = +5.0V in 50 seconds. Turning RH2-1C CCW increases the time ramp.

D. Minimum RUN ref. (elem. sh. AX and elem. sh. 310V).

Adjust R6-1D so that pin 30-1D = +3.8V. Turning R6-1D CS increases voltage.

8-2 Speed Feedback

A. Pulse Tachometer (elem. 1AY and elem. sh. 310S)

1. Verify the Berg jumpers (elem. sh. 310S).

2. With motor still, adjust R20-1L (NTMA-MTZA) until pin 2-1L = 0.0v.

3. For this part, the motor must be running. Run the motor on the 50% standby speed. Verify the motor RPM (N). The speed feedback at pin 2-1L = (N/900 X-10V). (For 50% speed RPM = 450, pin 2-1L = -5V). Ad just R9 CW to increase value. (In negative direction).

4. Stop Motor.

B. Speed Detect Points (elem. sh. 310P + T)

1. Turn R3-1D, R2-1E, R2-1F fully CCW.

2. Apply a (-) test ref to TP11-1E until TP15-1E = +5V and turn R3-1D cs until LED CPA-1D just lights.

3. Increase reference until TP15-1E = +5.9V and turn R2-1E CW until LED CPC-1E just lights.

4. Increase reference until TP15-1E = +7.4V and turn R2-1F CW until LED CPC-1F just lights. Remove reference.

C. Digital Speed Readout (elem. sh. Ax and elem. sh. 310R)

1. Turn R5-1D fully CCW.

2. With 0.0 volts observed at TP15-1E, the digital speed meter reads 000. Remove cover and adjust null resistor if necessary.

3. Apply a (-) test reference to TP11-1E until TP15-1E = +10.0V. Turn R5-1D CW until the digital speedmeter reads 900.

4. verify that TP15-1J =10V (remove speed indication). Remove reference.

8-3 Current Feedback

A. Input Scaling A-Bridge [and B-Bridge] (elem. sh. BC [BF] and elem. sh. 320G [320K])

1. Remove DTFA from XTFA or remove the burden resistors (MAY DO A+B bridge at same time). Can input test reference to both TP8-B1A + TP8-B1B. B-bridge part is in brackets [ ].

2. Input a test reference of 7.78 to TP8-B1A [B1B] (NFCC-IA) and adjust CFB-B1A [B1B] until TP1-B1A [B1B] (NFCC-CFB) = -7.5V.

3. Turn SUP-B1A [B1B] fully CCW and R4-A1G [R4-A1E] fully CW.

4. Increase the test reference to 9.24V and turn R4-A1G [R4-A1E] CCW until LED ACOC-B1E [B2E] (NFOD) just lights.

5. Increase the test reference to 10.8V and turn SUP-B1A [B1B] CW until LED SUP-B1A [B1B] just lights.

6. Remove the reference and reset the NFCC card.

B. Current Balance Regulator (NCCA) (elem. sh. BB and elem. sh. 320C)

1. Turn B1A-B2F CCW until B1-B1A = 0.0V.

2. Apply a test reference to TP8-B1A and TP8-B1B and adjust until CFB-B1A [B1B]= -7.5V. Verify TP ACB-B2F = +7.5V and TP IAV=B2F = -7.5V.

3. Adjust R1-B2F (NCCA) until CFBAV-B2F = -7.5V. Turning R1 CW increases voltage in the negative direction.

4. Verify JCA-5 = -7.5V, JCA-17 and JCA-18 = -7.5V. Remove reference.

C. Ammeter

1. With 0.0V on CFB-B1A and CFB-B1B, adjust the null on A-AM (A-inverter ammeter) and B-AM so that each reads 0A.

2. Apply a (+) reference to IA of B1A and B1B so that CFB-B1A and CFB-B1B = -7.5V.

3. Adjust the pot (loosen locking nuts) on the back of the ammeters so that each reads 1320A.

D. Thermal Model A-Bridge [B-Bridge] (elem. sh. BD [BG] and elem. sh. 320Q [320R]).

Note that the TP5 and TP6 of some NFOC cards are mis-labelled. TP5 should be labelled -TJ and TP6 labelled W/C.

1. Set RH1-A1B [A1C] to obtain Tamb (+6.0V) at TP5-B1D [B2D] (-TJ-NFOC).

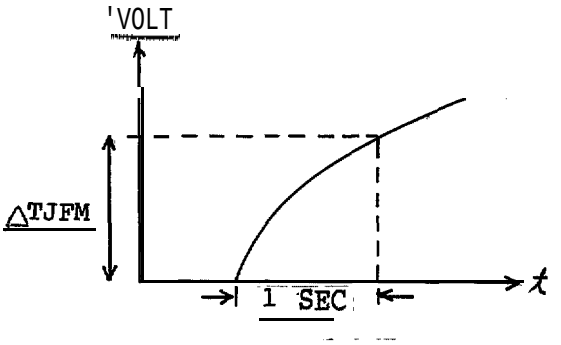
2. Connect DTM REF 1 to TP8-B1A [B1B] and adjust so that testpoint CFB-B1A [B1B] (NFOC) reads -7.5 volts. This value will be used for the calibration reference for CFB (FWD) in setting up the thermal model in the forward direction. Adjust R1-B1D [B2D] (NFOC) for V_{WL} (+3.57V) at TP6 (W/C). Turning R1 CW increases the output voltage.

3. Jumper pin 13-B1D [B2D] (NFOC) to ACOM (pin 9). Adjust R4-B1D [B2D] (NFOC) so that TP5 (-TJ) = -(Tamb-TJFC) = -4.72V. This voltage should be set at its steady-state value (may take several minutes to settle). Turning R4 CCW increases output in positive direction.

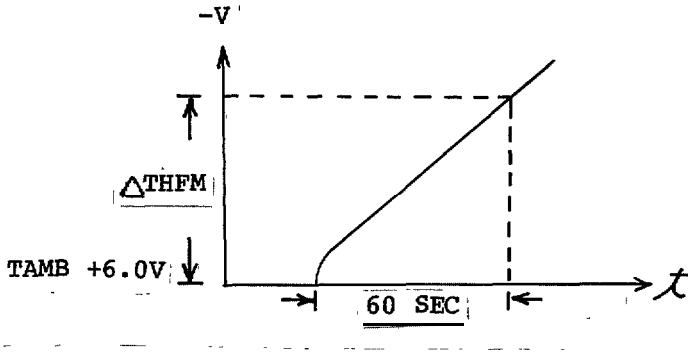
4. Remove the jumper from pin 13-B1D [B2D] (NFOC). Adjust R5-B1D [B2D] for THFC (-0.1V) at TP5 (-TJ). This voltage should be Set at its steady-state value. Turning R5 CCW increases output in positive direction.

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5. Jumper pin 13-B1D[B2D] (NFOC) to ACOM. Without disturbing REF 1 setting, open the reference switch and allow the voltage at TP5-B1D [B2D] (-TJ) to settle to 0.0 volts. Connect a chart recorder to TP5 (-TJ). Apply step reference with reference switch. On NFOC, adjust R2-B1D [B2D] △TJFM in 1 second as shown below. Note: Allow voltage at TP5 to settle to 0.0 volt before each step. Turning R2 CCW speeds up the response.



6. Remove jumper from pin 13-B1D [B2D] (NFOC) to ACOM. Leave chart recorder connected to TP5-B1D [B2D] (-TJ) without disturbing REF 1 setting. Open the reference switch and allow the voltage at TP5 to settle to +6.0 volts. To speed up the settling process, which can be quite slow, momentarily input volts into pin 16-B1D [B2D] (NFOC) to SA8 (DTM) [SA7 (DTM)] force TP5 to +6.1 volts. Apply step reference with reference switch. On NFOC adjust R3-B1D [B2D] for a voltage △THFM (2.2V) in 60 seconds as shown below. Note: Allow voltage at TP5 to settle to +6.0 volts before each step. Turning R3 CCW speeds up the response.



7. This completes the thermal modeling calibration.

8. Jumper pin 21 to 22 of A1G for inverter A, pin 23 to 24 of A1F for inverter B. (See sh. EE 320R). Adjust REF 1 for -12.5 volts at CFB. Light TOL and CTMP should light when TP5-B1D [B2D] reaches -7.6 volts. Remove REF 1, TOL light should latch in and CTMP light goes out, Press reset switch on NFCC board should turn off TOL.

8-4 Voltage Feedback (VFB)

A. Input Scaling (elem. sh. BC [BF] and elem. sh. 320H [320L])

1. Attach a Simpson between the P and N buses set to read at least 850 VDC and adjust null on voltmeter (VM) so that it reads 0V.

2. Start motor and transfer to KDS mode and apply minimum reference. Simpson should read 850 VDC.

3. Adjust VFB-B1A [B1B] so that TP13-B1A (NFCC-VFB) = -3.59V. Turning VFB CW increases the output.

4. Verify VFADV-B2F = -3.598 and continues with B.

B. Voltmeter (vm) (elem. sh. AX and elem. sh. 310R)

1. Adjust null when VFB-B1A and VFB-B1B = 0.0V so that Vm reads 0V.

2. Adjust RH 4-1C so that vm reads 850 volts when both VFB (NFCC cards) = -3.59V (see 8.4A). Turning RH4-1C CW increases Vm reading.

8-5 Rotor Volts (elem. sh. AW and elem. sh. 310T)

1. Turn R1-1D fully CW, turn RH3-1C fully CCW and R6-1E fully CW.

2. Simulate the rotor voltage at stall by applying a (+) ref to TP11-1B and adjusting until TP12-1B = -8.5V. Jumper pin 32-1E to pin 1-1E (simulates A2 closed) and turn R6-1E CW until TP12-1F = 0.0V.

3. Remove ref and turn RH3 CW until TP12-1F = -10V. Remove jumper.

4. When motor is running in RDS mode at 50% speed, turn R1-1D CCW until TP12-1F = -5v.

8-6 Speed/Volt Comparison (elem. sh. AW and elem. sh. 310N)

1. Turn R2-1D fully CW.

2. Apply a (-) ref to TP11-1E until TP15-1E = +5v and a second (+) ref to TP145-1F Until TP12-1F = -5.5V.

3. Turn R2-1D CCW until LED CPC-1D just lights.

4. Remove references and reset NFCC card.

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8-7 20% Current Detector (elem. sh. AW and elem. sh. 310P)

1. Turn R2-1G fully CW.

2. Apply a reference of 1.5V to TP14-1G and turn R2 CCW until LED CPC-1G just lights.

8-8 Shorted Diode Detector (elem. sh. AW and elem. sh. 310Q)

1. Turn RH5-1B fully CW and RH2-1B fully CW, R2-1G fully CW.

2. Turn RH2-1B CCW until pin 19-1B = +8.9V.

3. Apply a (-) reference to TP11-1D until TP15-1D = +8.5V, apply a second reference (+) to TP11-1B until TP12-1B = -4.25V and jumper pin 3 (P5V) to pin 74-1B. (Simulate 50% speed). Turn RH5-1B CCW until TP8-1B = 0.0v.

4. Reduce the second reference (TP11-1B) until TP12-1B = -3.8V. TP8-1B should = -1V. Turn R3-1G CCW until LED CPA-1G just lights.

5. Remove references and jumper and reset NFCC card if necessary.

8-9 Tach Preconditioning (elem. sh. AY and elem. sh. 310V)

1. Turn R5-1F fully CCW and jumper pin 65-16 to pin 66-16.

2. Apply a reference (+) to TP11-1F until TP15-1F = -5V. Turn R5 CW until TP15-16 = +2.5V.

3. Remove reference and jumper.

8-10 Current Regulation (elem. sh. AY and elem. sh. 310Y)

A. Preliminary Set-up

1. Adjust RH3-1B for +10V on pin 25-1B. turning RH3 CCW increases voltage.

2. See Chapter 6-7.

B. Operating Tuning

1. Run motor in KDS mode and do procedure in Chapter 6-7B. Adjust RH5-1C for proper response.

8-11 Speed Regulator (elem. sh. AY and elem. sh. 310X)

A. Preliminary Set-up

1. Adjust **R4-1K (NSCA-FWD** CL) until **TP1=-7.5V.**

2. Turn **R2-1K** fully CW.

B. Operational tune-up

See Chapter 6-8.5.

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